

Variance Estimation with Dependence and Heterogeneous Means

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Abstract

This paper develops a framework for variance estimation under dependence and heterogeneous means. This paper shows that consistent estimation of the variance target is impossible in general, and characterizes necessary and sufficient conditions for conservative variance estimation using dual cones. To choose among the valid estimators, this paper formulates three criteria — minimal correction, pointwise level estimand, and pointwise MSE — and shows how an eigenvalue truncation solution is optimal under all three criteria. This characterization and solution allow us to assess if existing variance estimators are valid and optimal in their respective settings, and construct the first optimal variance estimator that is simultaneously robust to heterogeneous means and cross-cluster serial correlation.

Keywords: variance estimation, heterogeneous means, two-way clustering, HAC, conic duality, design-based inference.

JEL codes: C12, C13, C23, C33.

1 Introduction

Suppose $Y_n \in \mathbb{R}^n$ is a vector of random variables with $E[Y_n] = \mu \in \mathcal{M}_n$ and $Var(Y_n) = \Sigma \in \mathcal{S}_n$. We are interested in a variance target $\theta_n = \text{tr}(W_n \Sigma)$ for some

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known real symmetric weight matrix W_n . The object θ_n can be interpreted as the oracle variance for $\sum_i Y_{n,i}$, which is used for inference after applying a central limit theorem. As an example, with independent observations, $W_n = I_n$, but W_n also encodes dependent observations such as one/ two- way clustering and serial correlation. To obtain the variance target, we rely on a variance estimator that takes a quadratic form $Y_n' A_n Y_n$, for some A_n . Observe that $E[Y_n' A_n Y_n] = \text{tr}(A_n \Sigma) + \mu' A_n \mu$. When the mean is known or consistently estimable, the second term $\mu' A_n \mu$ is innocuous and can be removed by demeaning, and the first term can recover the target. The difficulty confronted in this paper is when means can be arbitrarily heterogeneous, so the means are no longer consistently estimable. Considering heterogeneous means with a general dependence structure is the novel problem addressed in this paper.

Heterogeneous means arise in several settings. Under design-based uncertainty, where potential outcomes are fixed and cluster dependence reflects correlated assignment or sampling (Abadie et al., 2023), the score for each observation is heterogeneous and not consistently estimable by construction. In asset pricing, when we condition on the set of assets and time observed, any omitted priced factor leaves the regression score with a mean that varies across assets and time. In both cases, even though a central limit theorem governs the sum, the summands' means remain unidentified and the variance of the sum becomes difficult to estimate.

This paper makes three contributions.

First, this paper establishes the limits of identification in Section 3. With dependence and arbitrary mean heterogeneity, this paper shows that consistent estimation of the variance target $\theta_n = \text{tr}(W_n \Sigma)$ is impossible in general (Theorem 1), because the second moment alone is uninformative of the split between the covariance component and the mean component. Since consistency is unattainable, we can only pursue the second best goal, a conservative variance estimand, so that the associated statistical test remains valid (i.e., with size guarantee). Disparate estimators are quadratic forms governed by the same two objects: a linear subspace $\mathcal{M}_n$ of admissible mean vectors and a closed convex cone $\mathcal{S}_n$ of admissible covariance matrices. Necessary and sufficient conditions for conservativeness are derived (Theorem 2): $Y_n' A_n Y_n$ is uniformly conservative for θ_n if and only if $A_n - W_n$ lies in the dual cone $\mathcal{S}_n^*$ and A_n is positive semidefinite on $\mathcal{M}_n$. I then propose one such general-purpose variance estimator that sets A_n to be an eigenvalue truncation (ET) of W_n .

In light of how there are many variance estimators that are valid, the second

contribution of the paper is to show that ET is optimal in Section 4. Optimality is defined relative to criteria, so this paper proposes three natural criteria — minimal correction (which respects the dependence structure), minimum level of the variance estimand (in the interest of having the smallest variance used in the t-statistic and hence the most powerful test), and mean-squared error (MSE) (which is a common benchmark that assesses the quality of an estimator). Under some conditions, this solution is optimal under all three criteria. (Theorem 3)

Given the framework and conditions for validity and optimality, the third contribution of the paper is to assess if existing variance estimators in economically relevant settings with dependence are valid and optimal under heterogeneity, and if they are not, obtain the optimal conservative variance estimator. Section 5 considers several existing examples. The plug-in one-way cluster robust variance estimator is valid under mean heterogeneity (a result known from Abadie et al. (2023)), but now we also certify that it is optimal. In two-way clustering, the plug-in variance estimator from Cameron et al. (2011) (CGM) is not valid under mean heterogeneity, while the CGM2 modification from Davezies et al. (2021) is valid — these results are known from Xu and Yap (2024), and are corollaries of this paper’s setting. While CGM2 is valid, the new result is that CGM2 is not optimal as it does not match the ET solution — the difference is that CGM2 adds observation-specific variances to CGM, while ET adds the observation-specific within-transformed variances, which results in a smaller variance estimand. Section 6 extends the two-way cluster dependence to allow for cross-cluster weak dependence, a dependence structure considered in Chiang et al. (2024) (CHS). The plug-in variance estimator from CHS is invalid under mean heterogeneity, so a novel optimal conservative variance estimator is proposed as a corollary of this paper’s theory. This section also develops the ψ -dependence framework, the central limit theorem (Theorem 4), and consistency (Theorem 5). Simulations and a design-based empirical application on the impact of minimum wage policies close the section.

Related literature. This paper relates to three strands of literature.

The first is variance estimation in dependent arrays. Cameron et al. (2011) and Chiang et al. (2024) provide variance estimators under the two-way clustering and two-way clustering with cross cluster correlations respectively, but both assume mean homogeneity. This paper extends their variance estimators to be robust to mean heterogeneity. This paper also provides a central limit theorem that is weaker than

the one used in CHS, by relying on the ψ -dependence limit theory from [Kojevnikov et al. \(2021\)](#) (KMS) that does not rely on an Aldous-Hoover representation, thereby accommodating data-generating processes beyond the representation, similar to [Yap \(2025\)](#).

The second is distributional and mean heterogeneity. In time series, [Chan \(2022\)](#) and [Casini \(2023\)](#) permit time-varying means and second moments but require enough regularity to estimate the mean function consistently, whereas we impose no regularity on the mean sequence and instead modify the estimand to guarantee conservativeness. The interaction between dependence and mean heterogeneity has been rather disparate in design-based settings: settings with independence ([Abadie et al., 2020](#)) or one-way clustering ([Abadie et al., 2023](#)) have conservative plug-in variances, while two-way clustering does not ([Xu and Yap, 2024](#)). This paper’s framework unifies these dependence structures, and the conservativeness criterion gives a general and practical way to check if a dependence-robust plug-in is conservative. Further, this paper extends conservative variance estimation under mean heterogeneity beyond these settings to the CHS dependence structure, and derives optimal variances.

Third, a long tradition in time-series and clustered variance estimation is concerned that a quadratic form $Y_n'W_nY_n$ can return a negative number in finite samples when W_n is not positive semidefinite (PSD). [Newey and West \(1987\)](#) address this ex ante by adopting the Bartlett kernel; [Andrews \(1991\)](#) characterizes the class of kernels delivering positive-semidefinite estimates and identifies the quadratic-spectral kernel as minimizing asymptotic mean-squared error within it. A complementary ex post approach takes a possibly non-PSD estimate and projects it onto the PSD cone by eigenvalue correction, the operation which appears as the eigenvalue correction of [Chiang et al. \(2024\)](#). This paper’s framework differs in two respects. First, the non-PSD-ness we confront is not a sampling-noise artifact of a particular realization but a structural feature of the population weight matrix that the dependence design dictates so it does not vanish as the sample grows. Second, and consequently, our optimality is stated for the estimand, not the estimate: where the classical corrections minimize the distance of an estimate to the positive-semidefinite cone, ours minimizes the objective of the targeted variance subject to conservativeness.

2 Framework and Quadratic Representation

2.1 Statistical model and parameter

For each $n \geq 1$, let $Y_n = (Y_{n,1}, \dots, Y_{n,n})' \in \mathbb{R}^n$ be observed, distributed according to P_n . Define

$$\mu(P_n) := E_{P_n}[Y_n], \quad \Sigma(P_n) := \text{Var}_{P_n}(Y_n).$$

This paper will drop the indexing on P_n when it is without ambiguity.

The researcher commits to two structural objects:

- a linear subspace $\mathcal{M}_n \subseteq \mathbb{R}^n$ of admissible mean vectors;
- a closed convex cone $\mathcal{S}_n \subseteq \text{Sym}_n^+$ of admissible covariance matrices, where Sym_n^+ denotes the PSD cone in the space of symmetric $n \times n$ matrices Sym_n .

The model is

$$\mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n) := \{P_n : E_{P_n}\|Y_n\|^2 < \infty, \mu(P_n) \in \mathcal{M}_n, \Sigma(P_n) \in \mathcal{S}_n\}.$$

Let $W_n \in \text{Sym}_n$ be a known symmetric weight matrix. The parameter of interest is

$$\theta_n(P_n) := \text{tr}(W_n \Sigma(P_n)), \quad P_n \in \mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n).$$

An important context where θ_n is interesting is whenever some central limit theorem is applied while allowing for dependence across observations and mean heterogeneity. The central limit theorem yields $(\sum_{i=1}^n Y_{n,i} - E[Y_{n,i}]) / \sqrt{\text{Var}(\sum_{i=1}^n Y_{n,i})} \xrightarrow{d} N(0, 1)$. While $\frac{1}{n} \sum_{i=1}^n E[Y_{n,i}]$ is consistently estimable from the theorem, the individual means may not be when there is sufficient heterogeneity, which makes the estimation of $\text{Var}(\sum_{i=1}^n Y_{n,i})$ tricky when conducting statistical inference. When W_n corresponds to an adjacency matrix and encodes several dependence structures, this parameter of interest θ_n is asymptotically similar to $\text{Var}(\sum_{i=1}^n Y_{n,i})$ (i.e., $\text{Var}(\sum_{i=1}^n Y_{n,i}) / \theta_n = 1 + o(1)$) under some conditions, even when dependence across $Y_{n,i}$ is permitted. For instance, $W_n = I_n$ corresponds to independent observations. In time series settings with a kernel adjustment for the variance, W_n would reflect these kernel adjustments.

The pair $(\mathcal{M}_n, \mathcal{S}_n)$ encodes the researcher's structural commitments. Larger $\mathcal{M}_n$ and $\mathcal{S}_n$ correspond to weaker assumptions and stronger demands for robustness. Two extreme cases are instructive. At one end, $\mathcal{S}_n = \text{Sym}_n^+$ imposes no structure on the covariance and yields the maximally dependence-robust posture: the researcher seeks

conservativeness against any covariance compatible with $\Sigma \succeq 0$. At the other end, $\mathcal{S}_n = \{\sigma^2 I_n : \sigma^2 \geq 0\}$ encodes iid sampling and admits the classical demeaned sample variance. Intermediate cones encode block independence (clusters), bandedness (m -dependence), or diagonal structure (without correlation).

Examples of $\mathcal{S}_n$.

- $\mathcal{S}_n^{\text{iid}} := \{\sigma^2 I_n : \sigma^2 \geq 0\}$: iid sampling.
- $\mathcal{S}_n^{\text{diag}} := \{\Sigma \succeq 0 : \Sigma \text{ diagonal}\}$: heterogeneous variances but uncorrelated.
- $\mathcal{S}_n^{\text{block}} := \{\Sigma \succeq 0 : \Sigma_{ij} = 0 \text{ if } g(i) \neq g(j)\}$: cluster independence.
- $\mathcal{S}_n^{\text{band}(b)} := \{\Sigma \succeq 0 : \Sigma_{ij} = 0 \text{ if } |i - j| > b\}$: b -dependence.

Similarly, $\mathcal{M}_n$ can encode the known structure of the problem. Fully unrestricted means corresponds to $\mathcal{M}_n = \mathbb{R}^n$, while constant means corresponds to $\mathcal{M}_n = \text{span}\{1_n\}$, as two extreme cases.

2.2 Quadratic-form estimators and dual cones

A quadratic-form estimator takes the form $\hat{V}_{A,n}(Y_n) := Y_n' A_n Y_n$ for some $A_n \in \text{Sym}_n$, with expectation

$$V_{A,n}(P_n) := E_{P_n}[Y_n' A_n Y_n] = \text{tr}(A_n \Sigma(P_n)) + \mu(P_n)' A_n \mu(P_n). \quad (1)$$

$V_{A,n}(P_n)$ is an estimand, which is not necessarily the target $\theta_n(P_n)$.

The canonical plug-in estimator for the sum of Y_n (i.e., $1_n' Y_n$) corresponds to $A_n = W_n$, with bias

$$V_{W,n}(P_n) - \theta_n(P_n) = \mu(P_n)' W_n \mu(P_n).$$

This bias can be negative when W_n has negative directions inside $\mathcal{M}_n$, arising in anticonservativeness in inference.

To analyze conservativeness across the full pair $(\mathcal{M}_n, \mathcal{S}_n)$, we use cone duality. The *dual cone* of $\mathcal{S}_n$ is

$$\mathcal{S}_n^* := \{B \in \text{Sym}_n : \text{tr}(B\Sigma) \geq 0 \text{ for all } \Sigma \in \mathcal{S}_n\}. \quad (2)$$

Since $\mathcal{S}_n \subseteq \text{Sym}_n^+$ and Sym_n^+ is self-dual, $\mathcal{S}_n^* \supseteq \text{Sym}_n^+$, with equality when $\mathcal{S}_n = \text{Sym}_n^+$. For strict subcones, $\mathcal{S}_n^*$ contains symmetric matrices that are not PSD, and this larger

admissible region is precisely what licenses non-PSD corrections such as demeaning.

Examples of dual cones, all easily verified from (2):

$$(\mathcal{S}_n^{\text{iid}})^* = \{A : \text{tr}(A) \geq 0\}, \quad (\mathcal{S}_n^{\text{diag}})^* = \{A : \text{diag}(A) \geq 0\}.$$

For $\mathcal{S}_n^{\text{block}}$, $A \in \mathcal{S}_n^{\text{block},*}$ if and only if each within-cluster diagonal block of A is positive semidefinite, with arbitrary off-block entries. The covariance cones sit in a partial order: $\mathcal{S}_n^{\text{iid}} \subset \mathcal{S}_n^{\text{diag}} \subset \mathcal{S}_n^{\text{block}} \subset \text{Sym}_n^+$. Dual cones reverse the order.

3 Impossibility, Conservativeness, and Proposal

3.1 When consistent estimation is impossible

The same second moment matrix may admit different decompositions into a mean component and a covariance component:

$$E[Y_n Y_n'] = \Sigma(P_n) + \mu(P_n) \mu(P_n)'.$$

The target $\theta_n(P_n) = \text{tr}(W_n \Sigma(P_n))$, however, depends only on the covariance component. Hence, if the model permits a rank-one term $\mu \mu'$ to be shifted between covariance and mean while preserving admissibility, then the target changes but the expectation of every quadratic estimator does not. This yields an impossibility result for uniform consistency over the quadratic class.

Theorem 1 (Impossibility of consistency). *Fix n and consider the model $\mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n)$. Suppose there exist $\Sigma_0 \in \mathcal{S}_n$ and $v_n \in \mathcal{M}_n$ with $\Sigma_0 + v_n v_n' \in \mathcal{S}_n$. Then, for any $A_n \in \text{Sym}_n$,*

$$\sup_{P_n \in \mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n)} |\mathbb{E}_{P_n}[\widehat{V}_{A,n}] - \theta_n(P_n)| \geq \frac{1}{2} |v_n' W_n v_n|.$$

To gain some intuition for the result, suppose $\mathcal{S}_n$ is the set of diagonal matrices, and $v_n = (1, 0, \dots, 0)' \in \mathcal{M}_n$. Then, the second moment alone cannot distinguish $\mu_1 = 1$ and $\Sigma_{11} = 1$ from $\mu_1 = 0$ and $\Sigma_{11} = 2$. The proof rests on a two-point moment-matching argument: two distributions $P_n, Q_n \in \mathcal{P}_n$ can have the same expectation of every quadratic functional yet different values of θ_n , rendering them indistinguishable in expectation to the quadratic class.

As a corollary, write $\lambda_n \asymp \theta_n$ for the scale of the target. If $v_n' W_n v_n \neq o(\lambda_n)$, then the relative worst-case bias of every quadratic estimator is bounded below,

$$\frac{1}{\lambda_n} \sup_{P_n \in \mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n)} |\mathbb{E}_{P_n}[\widehat{V}_{A,n}] - \theta_n(P_n)| \geq \frac{|v_n' W_n v_n|}{2\lambda_n} \not\rightarrow 0.$$

Consequently, no quadratic estimator characterized by A_n is consistent in that $\widehat{V}_{A,n}/\theta_n \xrightarrow{p} 1$, uniformly over the model: consistency would force relative asymptotic unbiasedness $\mathbb{E}_{P_n}[\widehat{V}_{A,n}]/\theta_n \rightarrow 1$ at each point, contradicting the display.

Example 1 (Demeaned sample variance). The following example does not satisfy the conditions of the impossibility result as it does not contain sufficient heterogeneity in $\mathcal{M}_n$. Let $W_n = I_n$, $\mathcal{M}_n = \text{span}(\mathbf{1}_n)$, and $\mathcal{S}_n = \mathcal{S}_n^{\text{iid}}$, so $\theta_n = n\sigma^2$. Then $\mathcal{S}_n^* = \{A : \text{tr}(A) \geq 0\}$, and

$$A_n = \frac{n}{n-1} M_{\mathbf{1}_n}, \quad M_{\mathbf{1}_n} := I_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n'.$$

The estimator $Y_n' A_n Y_n = \frac{n}{n-1} \sum_i (Y_{n,i} - \bar{Y}_n)^2$ is exactly unbiased for $n\sigma^2$ on the model.

The impossibility in Theorem 1 requires the cone $\mathcal{S}_n$ to admit a perturbation along the mean direction: there must exist $\Sigma_0 \in \mathcal{S}_n$ and $v_n \in \mathcal{M}_n$ with $\Sigma_0 + v_n v_n' \in \mathcal{S}_n$ and $v_n' W_n v_n \neq 0$. In the iid case this feasibility condition fails. Any $v_n \in \mathcal{M}_n = \text{span}\{\mathbf{1}_n\}$ has the form $v_n = \alpha \mathbf{1}_n$, so $v_n v_n' = \alpha^2 \mathbf{1}_n \mathbf{1}_n'$, a rank-one matrix with nonzero off-diagonal entries. For any $\Sigma_0 = \sigma_0^2 I_n \in \mathcal{S}_n^{\text{iid}}$, the perturbed covariance $\Sigma_0 + \alpha^2 \mathbf{1}_n \mathbf{1}_n'$ is not a scalar multiple of I_n and therefore lies outside $\mathcal{S}_n^{\text{iid}}$ whenever $\alpha \neq 0$. The cone is too narrow for the adversary to hide mean variation inside a covariance perturbation, and consistent estimation of $\theta_n = n\sigma^2$ is restored.

3.2 Characterization of conservative estimands

In light of how consistent variance estimation may be impossible, we may use conservative variance estimators to avoid being oversized in inference. The following result provides necessary and sufficient conditions for a variance estimand to be conservative (and hence valid) using the dual cone representation.

Theorem 2 (Conservativeness via dual cones). *The estimand $V_{A,n}(P_n) = E_{P_n}[Y_n' A_n Y_n]$ satisfies $V_{A,n}(P_n) \geq \theta_n(P_n)$ for every $P_n \in \mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n)$ if and only if*

$$(C1) \quad A_n - W_n \in \mathcal{S}_n^*, \tag{3}$$

$$(C2) \quad \mu' A_n \mu \geq 0 \text{ for all } \mu \in \mathcal{M}_n. \quad (4)$$

In this paper, a variance estimand that satisfies (C1) and (C2) is considered *valid*, and a variance estimator is *valid* if its corresponding estimand is valid.

When $\mathcal{S}_n = \text{Sym}_n^+$, self-duality gives $\mathcal{S}_n^* = \text{Sym}_n^+$ and (C1) reduces to $A_n - W_n \succeq 0$ on $\mathbb{R}^n$, since $\text{tr}(B\Sigma) \geq 0$ for every PSD Σ if and only if B is itself PSD. For strict subcones, (C1) is weaker, admitting estimators that exploit covariance structure to sharpen the correction. A corollary of this lemma is that the plug-in variance estimator with W_n is conservative whenever W_n is positive semidefinite.

However, if $A_n = W_n$ is not positive semidefinite, (C1) can still be satisfied, and (C2) is still satisfied if $\mathcal{M}_n$ is sufficiently restricted (e.g., $\mathcal{M}_n = \{0\}$). Nonetheless, if $\mathcal{M}_n = \mathbb{R}^n$, then (C2) is equivalent to positive semidefiniteness of $A_n = W_n$, so W_n not being positive semidefinite implies that $V_{A,n}(P_n) < \theta_n(P_n)$ for some $\mu \in \mathcal{M}_n$. This result will be evident in CGM and m -dependence.

The characterization of conservativeness appears similar to the classical positive-semidefiniteness corrections in the variance-estimation literature, but the purpose and objective of the conservativeness are different. Corrections of the variance estimator are motivated by realized non-negativity: a quadratic form $Y_n' W_n Y_n$ with $W_n \not\succeq 0$ can return a negative number in a given sample, and the standard remedy enforces $\hat{V} \geq 0$ directly. A realized-non-negativity fix acts on the *estimate*: it adjusts the random variable $\hat{V}_{W,n}$ so that no draw falls below zero, but it does not change which estimand $V_{W,n}(P_n) = \text{tr}(W_n \Sigma) + \mu' W_n \mu$ is being targeted. When the plug-in is anticonservative, the problem is that $V_{W,n}(P_n) < \theta_n(P_n)$, i.e. $\mu' W_n \mu < 0$, even though $V_{W,n}(P_n)$ itself remains positive because the covariance term $\text{tr}(W_n \Sigma) = \theta_n$ dominates. The proposal developed in this paper acts on the *estimand*: replacing W_n with an A_n that satisfies the conditions of Theorem 2 changes the estimand.

3.3 The Eigenvalue Truncation Solution

The weight matrix W_n has the following spectral decomposition

$$W_n = U_n \Lambda_n U_n', \quad \Lambda_n = \text{diag}(\lambda_{n,1}, \dots, \lambda_{n,n}).$$

Write $\lambda^+ := \max\{\lambda, 0\}$ and $\lambda^- := \max\{-\lambda, 0\}$, set $\Lambda_n^\pm := \text{diag}(\lambda_{n,1}^\pm, \dots, \lambda_{n,n}^\pm)$, and define

$$W_n^+ := U_n \Lambda_n^+ U_n', \quad W_n^- := U_n \Lambda_n^- U_n', \quad \lambda_n^{\min} := \min_j \lambda_{n,j} = \lambda_{\min}(W_n),$$

so $W_n = W_n^+ - W_n^-$, both terms are positive semidefinite, and $\lambda_n^{\min}$ is the smallest eigenvalue of W_n . The *eigenvalue-truncation* (ET) weight matrix is:

$$A_n^\star := W_n^+ = U_n \Lambda_n^+ U_n'. \quad (5)$$

It is identical to W_n except that it truncates the negative eigenvalues at zero. $A_n^\star$ depends only on the known W_n ; it never references $\mathcal{S}_n$ and never requires an estimate of Σ . Due to Theorem 2, $A_n^\star$ is valid for any $(\mathcal{M}_n, \mathcal{S}_n)$, since W_n^+ is positive semidefinite.

Proposition 1 (When truncation is exact). *Let $A_n^\star = W_n^+$. For every $P_n \in \mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n)$,*

$$V_{A^\star, n}(P_n) - \theta_n(P_n) = \underbrace{\text{tr}(W_n^- \Sigma)}_{\text{covariance channel}} + \underbrace{\mu' W_n^+ \mu}_{\text{mean channel}}, \quad (6)$$

with $\text{tr}(W_n^- \Sigma) \geq 0, \mu' W_n^+ \mu \geq 0$. Further, $V_{A^\star, n}(P_n) = \theta_n(P_n)$ if and only if

$$\text{range}(\Sigma) \subseteq \ker W_n^- \quad \text{and} \quad \mu \in \ker W_n^+.$$

Besides stating the validity result formally, the proposition also provides necessary and sufficient conditions for the ET solution to be sharp in that it characterizes the set of (μ, Σ) such that the estimand is exact i.e., $V_{A^\star, n}(P_n) = \theta_n(P_n)$. The two conditions can be met without degeneracy: with $W_n = \text{diag}(1, -1)$, so that $A_n^\star = W_n^+ = \text{diag}(1, 0)$ and $W_n^- = \text{diag}(0, 1)$, any $\mu = c e_2$ with $c \neq 0$ and $\Sigma = \sigma^2 e_1 e_1'$ with $\sigma^2 > 0$ satisfies $\text{range}(\Sigma) = \text{span}(e_1) \subseteq \ker W_n^-$ and $\mu \in \ker W_n^+$, so $V_{A^\star, n}(P_n) = \theta_n(P_n) = \sigma^2$ even though $A_n^\star \neq W_n$.

4 Optimal Conservative Variance Estimation

4.1 Three optimization problems

By Theorem 2, every uniformly conservative quadratic estimator on $\mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n)$ corresponds to a matrix $A_n \in \text{Sym}_n$ satisfying (C1)–(C2). Among such estimators we seek one that is “best”. Three natural criteria arise: the first is a minimal correction criterion that minimizes the departure of A_n from W_n ; the second is a level criterion that minimizes the variance estimand, in the interest of the most powerful tests; and the third is the MSE, the common benchmark for the quality of an estimator.

The target $\theta_n = \text{tr}(W_n \Sigma)$ is a known function of W_n , and the plug-in $A_n = W_n$ estimates it with mean bias $\mu' W_n \mu$. When the plug-in is not conservative, the natural object is the one that repairs anticonservativeness while disturbing the plug-in as little as possible. We measure the disturbance by the worst-case upward adjustment of A_n relative to the plug-in W_n over admissible mean directions,

$$\mathcal{R}_n(A_n) := \sup_{\mu \in \mathcal{M}_n, \|\mu\|=1} \mu'(A_n - W_n)\mu. \quad (7)$$

Criterion 1: Minimal Correction. Let $P_{\mathcal{M}_n}$ denote the orthogonal projector onto $\mathcal{M}_n$. The minimal-correction problem is

$$\min_{A_n \in \text{Sym}_n} \mathcal{R}_n(A_n) \quad \text{subject to} \quad A_n - W_n \in \mathcal{S}_n^*, \quad P_{\mathcal{M}_n} A_n P_{\mathcal{M}_n} \succeq 0. \quad (8)$$

In the optimization problem (8) (C2) is imposed in the equivalent single-matrix form $P_{\mathcal{M}_n} A_n P_{\mathcal{M}_n} \succeq 0$. The equivalence is immediate: for every $\mu \in \mathcal{M}_n$ one has $P_{\mathcal{M}_n} \mu = \mu$, so $\mu' A_n \mu = \mu' P_{\mathcal{M}_n} A_n P_{\mathcal{M}_n} \mu$, and ranging over $\mu \in \mathcal{M}_n$ shows (C2) holds if and only if the restriction of A_n to $\mathcal{M}_n$ is positive semidefinite. The projected form is the usable one for computation: it is a single linear matrix inequality rather than an infinite family of scalar constraints.

Criterion 2: Level. The level of the estimand targeted by A_n is its expectation

$$V_{A,n}(\mu, \Sigma) = \mathbb{E}_{P_n}[Y_n' A_n Y_n] = \text{tr}(A_n \Sigma) + \mu' A_n \mu = \text{tr}(A_n(\Sigma + \mu \mu')),$$

and among conservative corrections one would like the one whose estimand is smallest, so as not to over-inflate the variance and lose power. The pointwise level problem at

a given admissible (μ, Σ) is

$$\min_{A_n \in \text{Sym}_n} \text{tr}(A_n(\Sigma + \mu\mu')) \quad \text{s.t.} \quad A_n - W_n \in \mathcal{S}_n^*, \quad P_{\mathcal{M}_n} A_n P_{\mathcal{M}_n} \succeq 0. \quad (9)$$

Criterion 3: MSE. With $\text{MSE}_n(A_n; \mu, \Sigma) := \mathbb{E}_{P_n}[(Y_n' A_n Y_n - \theta_n)^2]$, the pointwise MSE problem at a given admissible (μ, Σ) is

$$\min_{A_n \in \text{Sym}_n} \text{MSE}_n(A_n; \mu, \Sigma) \quad \text{s.t.} \quad A_n - W_n \in \mathcal{S}_n^*, \quad P_{\mathcal{M}_n} A_n P_{\mathcal{M}_n} \succeq 0. \quad (10)$$

Unlike (8), both (9) and (10) reference the unknown (μ, Σ) , and neither admits a worst-case formulation over the model.

Remark 1 (The bias and level criteria coincide). A natural objective to consider is to minimize the bias $|V_{A,n}(\mu, \Sigma) - \theta_n|$, but this problem coincides with the level problem. The bias of A_n decomposes as $V_{A,n}(\mu, \Sigma) - \theta_n = \text{tr}(A_n(\Sigma + \mu\mu')) - \text{tr}(W_n \Sigma)$, and $\text{tr}(W_n \Sigma)$ does not depend on A_n . Hence, minimizing the bias and minimizing the level are therefore the *same* optimization problem.

Remark 2 (A degenerate minimizer). Since (8) reads A_n only through its $\mathcal{M}_n$ -block, it cannot exclude trivial estimators. Take $\mathcal{S}_n^{\text{diag}}$, $\mathcal{M}_n = \mathbf{1}_n^\perp$, and $W_n = I_n$. Because $W_n = I_n$ acts as the identity on $\mathcal{M}_n$, every feasible A_n satisfies

$$\mathcal{R}_n(A_n) = \sup_{\mu \in \mathcal{M}_n, \|\mu\|=1} \mu'(A_n - I_n)\mu = \left(\sup_{\mu \in \mathcal{M}_n, \|\mu\|=1} \mu' A_n \mu \right) - 1 \geq -1,$$

the inequality by (C2), with equality if and only if $\mu' A_n \mu = 0$ for every $\mu \in \mathcal{M}_n$. The value -1 is therefore the minimum of (8), and it is attained by $A_n = \mathbf{1}_n \mathbf{1}_n'$, which is feasible — (C1) holds since $\text{diag}(\mathbf{1}_n \mathbf{1}_n' - I_n) = 0$, and (C2) since $P_{\mathcal{M}_n} \mathbf{1}_n = 0$ — and which yields the trivial estimator $\hat{V}_{A,n} = (\sum_i Y_{n,i})^2$. In contrast, the variance discipline of the MSE problem (10) rules such minimizers out.

4.2 Optimality of eigenvalue truncation solution

Throughout this subsection $\mathcal{M}_n = \mathbb{R}^n$, so that (4) reads $A_n \succeq 0$, and $\mathcal{S}_n$ is a closed convex cone. Fix an orthonormal eigenbasis $\{q_{n,j}\}_{j=1}^n$ of W_n , so that $W_n = \sum_{j=1}^n \lambda_{n,j} q_{n,j} q_{n,j}'$. The following definition is used for characterizing optimality.

Definition 1 (Aligned matrices). A matrix $A \in \text{Sym}_n$ is *aligned* if $A = \sum_j c_j q_{n,j} q'_{n,j}$ for some $c \in \mathbb{R}^n$; equivalently, if every $q_{n,j}$ is an eigenvector of A .

The definition applies to covariances and to weight matrices alike. Both W_n and $A_n^* = W_n^+$ are aligned. Optimality depends on whether $\mathcal{S}_n$ contains the eigenprojectors of W_n along which the plug-in is already conservative,

$$q_{n,j} q'_{n,j} \in \mathcal{S}_n \quad \text{for every } j \text{ with } \lambda_{n,j} > 0. \quad (11)$$

(11) delivers a set of scalar inequalities on the dual variable $A_n - W_n$, and we impose those directly, restricting the competitors against which A_n^* is compared. Let

$$\begin{aligned} \mathcal{S}_n^{*\oplus} &:= \{D \in \mathcal{S}_n^* : q'_{n,j} D q_{n,j} \geq 0 \text{ whenever } \lambda_{n,j} > 0\}, \\ \mathcal{A}_n^\oplus &:= \{A_n \in \text{Sym}_n : A_n - W_n \in \mathcal{S}_n^{*\oplus}, A_n \succeq 0\}. \end{aligned} \quad (12)$$

By construction $\mathcal{S}_n^{*\oplus} \subseteq \mathcal{S}_n^*$, so $\mathcal{A}_n^\oplus$ is contained in the uniformly conservative class on $\mathcal{P}_n(\mathbb{R}^n, \mathcal{S}_n)$. The inclusions are equalities if and only if (11) holds. Shrinking the dual cone enlarges its dual, so the primal cone becomes

$$\mathcal{S}_n^\oplus := (\mathcal{S}_n^{*\oplus})^* = \mathcal{S}_n + \text{cone}\{q_{n,j} q'_{n,j} : \lambda_{n,j} > 0\}, \quad (13)$$

the Minkowski sum of $\mathcal{S}_n$ with the cone generated by the eigenprojectors in question. By Theorem 2, $\mathcal{A}_n^\oplus$ is the class of weight matrices uniformly conservative on $\mathcal{P}_n(\mathbb{R}^n, \mathcal{S}_n^\oplus)$. Since $\mathcal{S}_n \subseteq \mathcal{S}_n^\oplus$, this is a stronger validity requirement than the researcher declared; the declared cone is what A_n^* is valid for, and $\mathcal{S}_n^\oplus$ is only the model at which its optimality is stated. Elements in $\mathcal{A}_n^\oplus$ have the following property:

$$q'_{n,j} A_n q_{n,j} \geq \lambda_{n,j}^+ \quad \text{for every } j \text{ and every } A_n \in \mathcal{A}_n^\oplus, \quad (14)$$

with equality throughout at $A_n = A_n^*$. Where $\lambda_{n,j} > 0$ this is the added inequality in (12); elsewhere it is $A_n \succeq 0$, since then $\lambda_{n,j}^+ = 0$.

The MSE criterion is a fourth-order object, so it cannot be evaluated from $(\mathcal{M}_n, \mathcal{S}_n)$ alone. We impose the following shape restriction, which makes the MSE a function of (μ, Σ) only.

Assumption 1 (No skewness, Gaussian fourth-cumulant structure). The distribution

P_n , with mean μ and covariance Σ , has finite fourth moments, and $Z := Y_n - \mu$ satisfies

$$\mathbb{E}_{P_n}[Z_i Z_j Z_k] = 0 \quad \text{for all } i, j, k, \quad \kappa_{ijkl}(P_n) = 0 \quad \text{for all } i, j, k, l,$$

where $\kappa_{ijkl}(P_n) = \mathbb{E}_{P_n}[Z_i Z_j Z_k Z_l] - \mathbb{E}[Z_i Z_j] \mathbb{E}[Z_k Z_l] - \mathbb{E}[Z_i Z_k] \mathbb{E}[Z_j Z_l] - \mathbb{E}[Z_i Z_l] \mathbb{E}[Z_j Z_k]$.

The no-skewness condition is implied by symmetry of Z about zero, $Z \stackrel{d}{=} -Z$; the Gaussian fourth-cumulant condition is implied by Gaussianity, and for elliptical distributions $\kappa^{(4)}$ is proportional to the Gaussian tensor scaled by a kurtosis parameter. Assumption 1 restricts the *shape* of Z at a given (μ, Σ) , not its location or scale, and is not a restriction to $\mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n)$.

Theorem 3 (Eigenvalue truncation: validity and optimality). *Assume $\mathcal{M}_n = \mathbb{R}^n$ and let A_n^* be as in (5). In (ii) and (iii), write $\Psi := \Sigma + \mu\mu'$ and $\psi_j := q'_{n,j} \Psi q_{n,j}$.*

(i) **Minimal correction.** *Suppose $\mathcal{S}_n \neq \{0\}$. Then, $A_n^* = W_n^+$ solves (8) with optimal value $\mathcal{R}_n(A_n^*) = (-\lambda_n^{\min})_+$.*

(ii) **Level.**

(a) Sufficiency. *For every aligned Ψ , $\text{tr}(A_n \Psi) \geq \text{tr}(A_n^* \Psi)$ for every $A_n \in \mathcal{A}_n^\oplus$, so A_n^* solves the level problem (9) posed at $\mathcal{S}_n^\oplus$. If $A_n \in \mathcal{A}_n^\oplus$ is aligned, then the inequality holds at every $\Psi \succeq 0$, whether or not Ψ is aligned.*

(b) Necessity. *If (11) fails at k , then some aligned A_n that is uniformly conservative on $\mathcal{P}_n(\mathbb{R}^n, \mathcal{S}_n)$ satisfies $\text{tr}(A_n \Psi) < \text{tr}(A_n^* \Psi)$ at $\Psi = q_{n,k} q'_{n,k}$, realized at the admissible pair $\Sigma = 0, \mu = q_{n,k}$.*

(c) Uniqueness. *A_n^* is the unique solution in (a) if and only if $\psi_j > 0$ for every j and the only $E \in \text{Sym}_n$ satisfying $q'_{n,j} E q_{n,j} = 0$ for every j and $A_n^* + E \in \mathcal{A}_n^\oplus$ is $E = 0$.*

(iii) **MSE.** *Let P_n satisfy Assumption 1 with aligned covariance Σ and mean μ proportional to $q_{n,i}$ for some i .*

(a) Sufficiency. *$\text{MSE}_n(A_n^*; \mu, \Sigma) \leq \text{MSE}_n(A_n; \mu, \Sigma)$ for every $A_n \in \mathcal{A}_n^\oplus$, so A_n^* solves the MSE problem (10) posed at $\mathcal{S}_n^\oplus$.*

(b) Necessity. *If (11) fails at k , then the A_n of (ii)(b) has strictly smaller MSE than A_n^* at the same pair $\Sigma = 0, \mu = q_{n,k}$.*

(c) Uniqueness. *A_n^* is the unique solution if and only if $\psi_j > 0$ for every j .*

Part (i) claims optimality on the minimal correction criteria that takes the worst-case μ . The conclusions of (ii) and (iii) are stronger in that they claim optimality

pointwise for (μ, Σ) , but the conditions they require are also more restrictive. This sufficiency result takes the intersection of dual cones that implies the containment of (11). If we were to solve the problem with the original primal cone $\mathcal{S}_n$, and (11) fails for the original primal cone, then (b) demonstrates how an alternative valid A_n can perform better than A_n^* , showing the necessity of that condition.

The level and MSE criteria receive largely similar treatment — the same class $\mathcal{A}_n^\oplus$, the same aligned second moments, and in (c) the same condition $\psi_j > 0$, which fails exactly when some eigendirection carries no second-moment mass. However, the MSE prices the off-diagonal entries of A_n in the eigenbasis, whereas the level $\text{tr}(A_n \Psi)$ at an aligned Ψ does not do so. Hence, the perturbations E appearing in (ii)(c) are not penalized by level criterion, but are penalized by the MSE criterion, so (iii)(c) needs no further condition on $\mathcal{S}_n$. Whether that second condition binds is a property of the declared cone: at $\mathcal{S}_n = \text{Sym}_n^+$, (3) forces $A_n - W_n \succeq 0$, whose eigenbasis diagonal vanishes wherever $\lambda_{n,j} > 0$, so $E = 0$; under a zero-pattern cone the ties survive.

The restriction to $\mathcal{M}_n = \mathbb{R}^n$ is the restriction to A_n being positive semidefinite. If A_n is not positive semidefinite, then if the support of Y is $\mathbb{R}^n$, then it is possible to obtain some data draw y such that $y' A_n y < 0$ so we have a negative variance estimate. To the extent that we want to exclude estimators that can give negative variance estimates, $\mathcal{M}_n = \mathbb{R}^n$ is reasonable.

The ET variance estimator is shown to converge to the intended variance estimand under some primitive conditions in Lemma 2 in the appendix. Hence, when coupled with a central limit theorem, the variance estimator can be used for robust and optimal inference. These limit theorems are often derived under particular dependence structures, so the formal statement is deferred to Section 6. We fix $A_n^* = W_n^+$ as the closed-form benchmark.

Remark 3. Outside the optimality environment of Theorem 3, one may instead use a weighted average of (μ, Σ) to construct an averaged criterion on levels and MSE, then numerically implement a conic optimization problem. While there are established algorithms for such optimizations, the resulting estimator is not guaranteed to converge to its estimand in general. Further discussion is left to Appendix A.1.

5 Examples and Optimal Corrections

5.1 One-way cluster-robust estimators

With observations partitioned into clusters $G_1, \dots, G_{C_n}$ and indicator vectors $B_c \in \mathbb{R}^n$, the one-way cluster-robust estimator is

$$\widehat{V}_{\text{clu},n} = \sum_{c=1}^{C_n} \left(\sum_{i \in G_c} Y_{n,i} \right)^2 = Y_n' W_n^{\text{clu}} Y_n, \quad W_n^{\text{clu}} = \sum_{c=1}^{C_n} B_c B_c'.$$

Each $B_c B_c'$ is positive semidefinite, so W_n^{clu} is globally PSD, and the plug-in is uniformly conservative for every admissible mean space. Hence $A_n^* = W_n^{\text{clu}}$, and heterogeneous means cannot induce anticonservativeness when the weight matrix is globally positive semidefinite. This result shows how the plug-in one-way cluster robust variance estimator is not only conservative in design-based settings (as established in [Abadie et al. \(2023\)](#)), it is also optimal.

5.2 Time series: HAC estimators

A generic HAC estimator is the quadratic form

$$\widehat{V}_{\text{HAC},n} = \sum_{i=1}^n \sum_{j=1}^n K\left(\frac{i-j}{b_n}\right) Y_{n,i} Y_{n,j} = Y_n' W_n^{\text{HAC}} Y_n, \quad (W_n^{\text{HAC}})_{ij} = K\left(\frac{i-j}{b_n}\right),$$

with W_n^{HAC} symmetric Toeplitz. Whether the plug-in $A_T^* = W_n^{\text{HAC}}$ is already optimal turns entirely on the *spectral window* of the kernel. For a symmetric banded Toeplitz matrix B with value c_k on its k -th diagonal, the spectral window is the generating function

$$f(\omega) := c_0 + 2 \sum_{k \geq 1} c_k \cos(k\omega), \quad \omega \in [-\pi, \pi],$$

the Fourier transform of the kernel weights. Two facts connect f to the eigenvalues of B : every eigenvalue lies in the range of f (each Rayleigh quotient is a weighted average of f values), and by Grenander–Szegő the empirical eigenvalue distribution converges to that of f . The sign of f therefore governs whether the plug-in is conservative. We contrast two kernels.

5.2.1 Naive m-dependence plug-in

Consider a scalar time series $\{Y_t\}_{t=1}^T$ that is m -dependent, so $\text{Cov}(Y_t, Y_{t+k}) = 0$ for $|k| > m$. The variance of the sum is

$$\text{Var}\left(\sum_{t=1}^T Y_t\right) = \sum_t \text{Var}(Y_t) + \sum_{k=1}^m \sum_t [\text{Cov}(Y_t, Y_{t+k}) + \text{Cov}(Y_{t+k}, Y_t)].$$

The naive plug-in replaces each covariance with the product $Y_t Y_{t+k}$ and weights all admissible lags equally,

$$\widehat{V}_T^{\text{naive}} = \sum_t Y_t^2 + \sum_{k=1}^m \sum_t (Y_t Y_{t+k} + Y_{t+k} Y_t) = Y' B_m Y,$$

where $B_m \in \text{Sym}_T$ is the symmetric banded Toeplitz matrix with ones on bands $0, 1, \dots, m$. Its spectral window is the Dirichlet kernel of order m ,

$$f_{\text{Dir}}(\omega) = 1 + 2 \sum_{k=1}^m \cos(k\omega) = \frac{\sin((m + \frac{1}{2})\omega)}{\sin(\omega/2)},$$

which oscillates in sign, vanishing at $\omega = 2\pi j/(2m + 1)$, $j = 1, \dots, m$, and taking negative values between successive zeros. A positive fraction of the eigenvalues of B_m are negative, so B_m is indefinite. Due to Theorem 2, the variance estimand can be anticonservative. The eigenvalue truncation $A_T^* = B_m^+$ addresses both issues.

The optimal correction is the positive-part operator

$$A_T^* = B_m^+ = U \max(\Lambda, 0) U',$$

where $B_m = U \Lambda U'$ is the spectral decomposition. In the frequency domain, the optimal estimator has spectral window $f^*(\omega) = \max\{f_{\text{Dir}}(\omega), 0\}$, modifying B_m only on the frequency band where $f_{\text{Dir}} < 0$.

5.2.2 Newey–West with Bartlett kernel

The Newey–West estimator replaces uniform weights by linearly declining Bartlett weights $\omega(k, M) = 1 - k/(M + 1)$ for $|k| \leq M$,

$$\widehat{V}_T^{\text{NW}} = Y' C_M Y, \quad (C_M)_{ts} = \omega(|t - s|, M),$$

with $C_M \in \text{Sym}_T$ banded Toeplitz. Its spectral window is the Fejér kernel,

$$f_{\text{Fej}}(\omega) = \frac{1}{M+1} \left(\frac{\sin((M+1)\omega/2)}{\sin(\omega/2)} \right)^2 \geq 0.$$

Since every eigenvalue of a symmetric Toeplitz matrix lies in the range of its spectral window, nonnegativity of f_{Fej} forces $C_M \succeq 0$ exactly, in every finite sample—the positive-semidefiniteness for which the Bartlett kernel was introduced (Newey and West, 1987). No truncation is required: $A_T^* = C_M$, and the identification problem of this paper does not bind in the Bartlett–HAC setting.

5.3 Two-way clustering

We work in a balanced panel with G groups and T periods, one observation per cell, so $n = GT$, and identify $\mathbb{R}^n \cong \mathbb{R}^T \otimes \mathbb{R}^G$. With row clusters indexed by g and column clusters by t , the Cameron–Gelbach–Miller (CGM) weight matrix is

$$W_n^{2\text{-way}} = W_n^G + W_n^T - W_n^{G \cap T}, \quad W_n^G = J_T \otimes I_G, \quad W_n^T = I_T \otimes J_G, \quad W_n^{G \cap T} = I_n,$$

where $J_G = \mathbf{1}_G \mathbf{1}_G'$ and $J_T = \mathbf{1}_T \mathbf{1}_T'$. Each component is positive semidefinite, but the subtraction renders $W_n^{2\text{-way}}$ indefinite, so under heterogeneous means the plug-in is anticonservative (Xu and Yap, 2024). Let

$$P_{00} := I_n - \frac{W_n^G}{T} - \frac{W_n^T}{G} + \frac{J_n}{n}, \quad J_n = \mathbf{1}_n \mathbf{1}_n',$$

be the orthogonal projector implementing two-way demeaning, $(P_{00} y)_{gt} = y_{gt} - \bar{y}_{\cdot t} - \bar{y}_{g \cdot} + \bar{y}_{\cdot \cdot}$, and the associated subspace $\mathcal{S}_{00} := \text{range}(P_{00})$.

Corollary 1 (Optimal two-way correction). *Under $\mathcal{M}_n = \mathbb{R}^n$, the eigenvalue-truncation estimator of Theorem 3 is*

$$A_n^* = (W_n^{2\text{-way}})^+ = W_n^{2\text{-way}} + P_{00},$$

equivalently $Y_n' A_n^* Y_n = Y_n' W_n^{2\text{-way}} Y_n + \sum_{g,t} (Y_{gt} - \bar{Y}_{\cdot t} - \bar{Y}_{g \cdot} + \bar{Y}_{\cdot \cdot})^2$.

The CGM matrix subtracts $W_n^{G \cap T} = I_n$ to undo the double-counting of cells entering both the row-cluster and column-cluster sums. Consequently, on the two-way within subspace (vectors v orthogonal to every row and column mean), we have

$W_n^{2\text{-way}}v = -v$, resulting in a negative eigenvalue of -1 . Truncating its eigenvalue to zero adds exactly the projector P_{00} onto that subspace. Since the within subspace is precisely where heterogeneous means generate anticonservativeness (Theorem 2), the correction places nonnegative mass only where the plug-in was deficient.

CGM2 uses weight matrix $W_n^G + W_n^T = A_n^* + (I_n - P_{00})$. Since $I_n - P_{00} \succeq 0$, CGM2 is uniformly conservative but non-optimal: it overshoots A_n^* on the between-group and between-time directions, where no correction is needed.

5.4 Two-way clustering with serial correlation

We now allow cross-cluster serial correlation on the time dimension. In the balanced panel ($n = GT$), the Chiang–Hansen–Sasaki (CHS) plug-in is

$$W_n^{\text{CHS}} = J_T \otimes I_G + C_M \otimes (J_G - I_G),$$

where $C_M \in \mathbb{R}^{T \times T}$ is the Bartlett kernel matrix, $(C_M)_{ts} = \omega(|t - s|, M)$ with $\omega(m, M) = \max\{1 - m/(M + 1), 0\}$. This result is stated in Lemma 1 of Appendix A. Let $P_G := J_G/G$ and $Q_G := I_G - P_G$ be the group-mean and group-demeaning projectors, and define the indefinite $T \times T$ block

$$B := J_T - C_M, \quad B^- := (C_M - J_T)_+ \succeq 0.$$

Under $\mathcal{M}_n = \mathbb{R}^n$, $A_n^* = (W_n^{\text{CHS}})^+$ by Theorem 3. Due to Theorem 2, the naive CHS variance estimator can be anticonservative, as exemplified in Appendix A.

Corollary 2 (Optimal correction under serial correlation).

$$A_n^* = (W_n^{\text{CHS}})^+ = W_n^{\text{CHS}} + B^- \otimes Q_G.$$

With $\check{Y}_{g,t} := Y_{g,t} - \bar{Y}_{\cdot t}$ the group-demeaned (cross-sectionally centered) score,

$$Y_n' A_n^* Y_n = Y_n' W_n^{\text{CHS}} Y_n + \sum_{g=1}^G \check{Y}_{g,\cdot}' B^- \check{Y}_{g,\cdot}. \quad (15)$$

Split the cross-section into the group-mean direction P_G and its complement Q_G ; the CHS weight acts block-diagonally across them. On the group-symmetric block the weight is $J_T + (G - 1)C_M$, a nonnegative combination of two PSD matrices, hence

already conservative. On the group-demeaned block the weight is $J_T - C_M$: the rank-one temporal aggregator J_T , concentrated on the constant-in-time direction, is partly canceled by the Fejér kernel C_M , producing negative eigenvalues. Only this block needs repair, and the truncation adds back precisely its negative eigenmass B^- , supported on the group-demeaned scores. The correction is thus a single $T \times T$ eigendecomposition applied within clusters. At $M = 0$, $C_M = I_T$ and $B^- = I_T - J_T/T$ is the time-demeaning projector, so $B^- \otimes Q_G$ reduces to two-way demeaning and we recover Corollary 1.

Example 2 (Exactness). Here $W_n^- = B^- \otimes Q_G$, so $\text{range}(W_n^-) = \text{range}(B^-) \otimes \text{range}(Q_G)$: the anticonservative directions are group-demeaned in the cross-section and load on the negative eigenspace of $B = J_T - C_M$. Proposition 1 is met by

$$\Sigma = \Sigma_T \otimes J_G, \quad \Sigma_T \succeq 0 \text{ with } (\Sigma_T)_{ts} = 0 \text{ for } |t - s| > M,$$

$$\mu = p \otimes u, \quad Bp = \beta p \text{ with } \beta \leq 0, \quad u \in \text{range}(Q_G).$$

The covariance is admissible, $\Sigma \in \mathcal{S}_n^{\text{CHS}}$, and describes a panel whose entire cross-sectional dependence is a common time factor with M -banded autocovariance. Then $(B^- \otimes Q_G)\Sigma = (B^- \Sigma_T) \otimes (Q_G J_G) = 0$, and since $A_n^* = [J_T + (G-1)C_M] \otimes P_G + B^+ \otimes Q_G$ we have $A_n^* \mu = (B^+ p) \otimes u = 0$. Both channels vanish and $V_{\text{ET}}^* = \theta_n$, so by Corollary 3 the t -test has asymptotic size exactly α : the size guarantee binds, and no uniformly conservative competitor improves on it.

Remark 4 (EVC floors the estimate; ET repairs the estimand). Chiang et al. (2024) render their plug-in non-negative through an eigenvalue correction (EVC) that replaces the negative eigenvalues of the *realized* meat $\hat{\Omega}_{NT}$ by zero. In the notation here this acts on the data-space realization $Y_n' W_n^{\text{CHS}} Y_n$. At the plug-in covariance the bias-free part $\text{tr}(W_n^{\text{CHS}} \Sigma_n) > 0$, so the estimand $V_{\text{CHS}} = \text{tr}(W_n^{\text{CHS}} \Sigma_n) + \mu_n' W_n^{\text{CHS}} \mu_n$ is positive and $\hat{\Omega}_{NT} \rightarrow V_{\text{CHS}} > 0$, leaving the correction asymptotically inactive—yet $V_{\text{CHS}} < V_{\text{true}}$ whenever $\mu_n' W_n^{\text{CHS}} \mu_n < 0$ (the counterexample of Appendix A).

6 Inference under Panel Dependence

6.1 Dependence framework

Consider a two-way cluster dependent setting, where one dimension exhibits cross-cluster weak dependence. To keep the exposition concrete, suppose we have a panel with both a cross sectional and a time series component. The time series component contains time indices $t \in \{1, 2, \dots, T\}$ for every observation, and observations within the same time period may be arbitrarily correlated. The cross sectional component can be partitioned on some dimension G , with cluster indices $g \in \{1, 2, \dots, G\}$ such that every observation within the same g cluster may be arbitrarily correlated. Let $i \in \{1, 2, \dots, n\} =: N_n$ index observations, $g(i)$ denote its cross sectional cluster, and $t(i)$ denote its time period. For example, in a canonical balanced panel with each cross sectional unit observed in every time period, we have $i = (g, t)$. While the setting in this paper permits unbalanced panels, the balanced panel with $G \asymp T$ is used as a running example.

There is a triangular array of multivariate random vectors $\{Y_{n,i}\}, n \geq 1, Y_{n,i} \in \mathbb{R}^v$, where n denotes the number of observations. Define the moment of this random vector as $\|Y_{n,i}\|_p = (E[\|Y_{n,i}\|^p])^{1/p}$. All observations i may be serially correlated. The distance $d_n(i, j)$ between any two observations i and j is 0 if they share a cluster on any dimension, and the difference in their time index otherwise.

Let N_t^T denote the set of observations in time t , N_g^G denote the set of observations in cluster g , and $N_{t,g}^{T \cap G} := N_t^T \cap N_g^G$ the set of observations in both time t and cluster g . Define the following objects to measure how concentrated observations are across clusters and time:

$$\delta_n^\partial(s; k) := \frac{1}{n} \sum_{i \in N_n} (|N_{t(i)+s}^T| + |N_{t(i)-s}^T| + \mathbf{1}\{s = 0\}(|N_{g(i)}^G| - |N_{t(i)}^T|))^k; \quad (16)$$

$$\Delta_n(s, m; k) := \frac{1}{n} \sum_{i \in N_n} \left(\sum_{h=s}^m (|N_{t(i)+h}^T| + |N_{t(i)-h}^T| + \mathbf{1}\{h = 0\}(|N_{g(i)}^G| - |N_{t(i)}^T|)) \right)^k \quad (17)$$

$$c_n(s, m; k) := \inf_{\alpha > 1} [\Delta_n(s, m; k\alpha)]^{1/\alpha} [\delta_n^\partial(s; \frac{\alpha}{\alpha-1})]^{1-1/\alpha}. \quad (18)$$

These objects can be interpreted in the following manner. $\delta_n^\partial(s; k)$ tells us the average number of observations that are s periods away, while $\Delta_n(s, m; k)$ tells us the number

of observations between m and s periods away (inclusive), both raised to the power of k . Finally, $c_n(s, m; k)$ gives us an upper bound on the convolution of $\delta_n^\partial(s; k)$ and $\Delta_n(s, m; k)$ when Hölder's inequality is applied, which is interpretable as a measure of how quickly neighborhoods grow and overlap. In the running example of a balanced panel, $c_n(s, m; k) \asymp (m - s)^k G^k$.

The rest of this section characterizes ψ -dependence, which is the sense in which observations are weakly dependent across clusters and across time. Let $\mathcal{L}_{v,a}$ denote the collection of bounded Lipschitz real functions on $\mathbb{R}^{v \times a}$:

$$\mathcal{L}_{v,a} = \{f : \mathbb{R}^{v \times a} \rightarrow \mathbb{R} : \|f\|_\infty < \infty, \text{Lip}(f) < \infty\}, \quad (19)$$

where $\text{Lip}(f)$ is the Lipschitz constant of f and $\|\cdot\|_\infty$ is the sup-norm. For any positive integers a, b, s , let $\mathcal{Q}_n(a, b; s)$ denote the collection of all pairs of observation sets of sizes a and b at distance at least s :

$$\mathcal{Q}_n(a, b; s) = \{(A, B) : A, B \subset N_n, |A| = a, |B| = b, \text{and } d_n(A, B) \geq s\}, \quad (20)$$

where $d_n(A, B) = \min_{i \in A} \min_{j \in B} d_n(i, j)$.

Inference is conditional on a σ -field $\mathcal{C}_n$ that holds fixed the information common to the array—for instance the cluster- and time-level common factors, or, in the design-based reading, the fixed potential outcomes. All expectations, variances, and covariances in this section and Appendix A.4 are taken conditional on $\mathcal{C}_n$, and the dependence coefficients θ_n below are $\mathcal{C}_n$ -measurable. Write $\mu_n := \mathbb{E}[Y_n \mid \mathcal{C}_n]$ and $\Sigma_n := \text{Var}(Y_n \mid \mathcal{C}_n)$ for the conditional first and second moments; these are the $\mathcal{C}_n$ -measurable analogues of the μ, Σ of Section 2. The dependence coefficients $\{\theta_{n,s}\}_{s \geq 0}$ below (notation following KMS) are unrelated to the scalar variance target $\theta_n = \text{tr}(W_n \Sigma)$ of Section 2, which does not recur in this section.

Definition 2 (ψ -dependence). A triangular array $\{Y_{n,i}\}, n \geq 1, Y_{n,i} \in \mathbb{R}^v$ is conditionally ψ -dependent if for each $n \in \mathbb{N}$, there exist a measurable sequence $\theta_n = \{\theta_{n,s}\}_{s \geq 0}, \theta_{n,0} = 1$, and a collection of nonrandom functions $(\psi_{a,b})_{a,b \in \mathbb{N}}, \psi_{a,b} : \mathcal{L}_{v,a} \times \mathcal{L}_{v,b} \rightarrow [0, \infty)$, such that for all $(A, B) \in \mathcal{Q}_n(a, b; s)$ with $s > 0$ and all $f \in \mathcal{L}_{v,a}$ and $g \in \mathcal{L}_{v,b}$,

$$|\text{Cov}(f(Y_{n,A}), g(Y_{n,B}) \mid \mathcal{C}_n)| \leq \psi_{a,b}(f, g) \theta_{n,s} \quad \text{a.s.} \quad (21)$$

where $Y_{n,A} := (Y_{n,i})_{i \in A}$. We call the sequence θ_n the dependence coefficients of $\{Y_{n,i}\}$.

This ψ -dependence setting generalizes the more familiar strong-mixing processes used in the two-way cluster dependence panel literature with serial correlation.

The setup here generalizes the CHS setting that requires a representation where $Y_{gt} = f(\alpha_g, \gamma_t, \varepsilon_{gt})$, for iid $\alpha_g, \varepsilon_{gt}$ and γ_t strictly stationary, which generalizes a standard Aldous–Hoover representation that additionally requires γ_t iid. An example that satisfies the conditions of this paper and has homogeneous means but does not permit the above representation is given in Appendix A.2.

6.2 Central limit theorem

The results are stated in a setting where $\sum_{i \in N_n} E[Y_{n,i}] = 0$, and we are interested in constructing the variance of $\sum_{i \in N_n} Y_{n,i}$. With $V_{\text{true}} := \text{Var}(\sum_{i \in N_n} Y_{n,i})$, let $\lambda_n := \lambda_{\min}(V_{\text{true}})$ denote its smallest eigenvalue. Since CHS uses an Aldous–Hoover representation, there is no mean heterogeneity, so their central limit theorem is incompatible with the setting of this paper. Consequently, a stronger central limit theorem that permits such heterogeneity is obtained in this subsection.

Assumption 2. The triangular array $\{Y_{n,i}\}$ is ψ -dependent with the dependence coefficients $\{\theta_n\}$ satisfying: (a) For some constant $C > 0$, $\psi_{a,b}(f, g) \leq C \times ab(\|f\|_\infty + \text{Lip}(f))(\|g\|_\infty + \text{Lip}(g))$; (b) $\sup_{n \geq 1} \max_{s \geq 1} \theta_{n,s} < \infty$; (c) For some $p > 4$, $\sup_{n \geq 1} \max_{i \in N_n} \|Y_{n,i}\|_p < \infty$ a.s.

Assumption 3. There exists a positive sequence $m_n \rightarrow \infty$ such that for $k = 1, 2$, and the p in Assumption 2,

- (a) $\frac{n}{\lambda_n^{1+k/2}} \sum_{s \geq 0} c_n(s, m_n; k) \theta_{n,s}^{1-(2+k)/p} \rightarrow 0$; and
- (b) $\frac{n^2 \theta_{n,m_n}^{1-(1/p)}}{\lambda_n^{1/2}} \rightarrow 0$.

These conditions balance the dependence strength, neighborhood growth, and variance growth. In an AR(1) process with positive coefficient ρ , $\lambda_n \asymp T$, and both conditions are satisfied with $m_n = T^{1/3}$ as $T \rightarrow \infty$. In the running balanced panel example with common random effects within clusters and an AR(1) process in the common time effects, $\lambda_n \asymp T^2 G$, and again $m_n = T^{1/3}$ suffices.

Theorem 4. Suppose Assumptions 2 and 3 hold a.s. Let ν be a nonstochastic vector with $\dim(\nu) = \dim(Y_{n,i})$ and $\|\nu\| = 1$. Then, for $S_n := \sum_{i \in N_n} \nu'(Y_{n,i} - E[Y_{n,i}])$ and $\sigma_n^2 := \text{Var}(S_n)$,

$$\sup_{t \in \mathbb{R}} \left| \Pr \left\{ \frac{S_n}{\sigma_n} \leq t \right\} - \Phi(t) \right| \xrightarrow{a.s.} 0, \quad (22)$$

where Φ denotes the distribution function of $N(0, 1)$.

This result suffices for a uniform multivariate central limit theorem due to the Cramer–Wold device.

6.3 Consistency

By Corollary 2, the optimal variance estimator is the eigenvalue truncation given in (15). The corresponding estimand is

$$V_{\text{ET}}^* := \mathbb{E}[\widehat{V}_{\text{ET}}^* \mid \mathcal{C}_n] = V_{\text{CHS}} + \sum_{g=1}^G \mathbb{E}[\check{Y}'_{n,(g,\cdot)} B^- \check{Y}_{n,(g,\cdot)} \mid \mathcal{C}_n]. \quad (23)$$

Using $y_t := \sum_{i \in N_t^T} Y_{n,i}$, we compare the estimand to the kernel-adjusted variance

$$\begin{aligned} V_{\text{adj}} := & \sum_{i \in N_n} \sum_{j \in N_{g(i)}^G} \text{Cov}(Y_{n,i}, Y_{n,j}) + \sum_{i \in N_n} \sum_{j \in N_{t(i)}^T} \text{Cov}(Y_{n,i}, Y_{n,j}) - \sum_{i \in N_n} \sum_{j \in N_{t(i),g(i)}^{T \cap G}} \text{Cov}(Y_{n,i}, Y_{n,j}) \\ & + \sum_{m=1}^M \sum_{t=1}^{T-m} \omega(m, M) [\text{Cov}(y_t, y_{t+m}) + \text{Cov}(y_{t+m}, y_t)], \end{aligned} \quad (24)$$

which will be shown to be asymptotically equivalent to V_{true} . This V_{adj} is $\text{tr}(W_n^{\text{CHS}} \Sigma_n)$, the truncated kernel covariance: the variance part of the CHS plug-in ($\mathbb{E}[\widehat{V}_{\text{CHS}}] = V_{\text{adj}} + \mu'_n W_n^{\text{CHS}} \mu_n$), and the quantity $\widehat{V}_{\text{ET}}^*$ targets.

We state the assumptions for the consistency theorem.

Assumption 4. $\frac{n}{\lambda_n^2} \sum_{s \geq 0} c_n(s, M; 2) \theta_{n,s}^{1-4/p} \rightarrow 0$.

This assumption mimics Assumption 4.1(iii) in KMS. In the balanced panel the left-hand side is $\asymp M^2/T^2$, so the condition reduces to $M = o(T)$; the choice $M \asymp T^{1/3}$ used in the simulations and application satisfies it.

Assumption 5. The following hold almost surely:

- (a) $\frac{n}{\lambda_n} \sum_{m=1}^M |\omega(m, M) - 1| \delta_n^\partial(m; 1) \theta_{n,m}^{1-2/p} \rightarrow 0$;
- (b) $\frac{1}{\lambda_n} \sum_{m \geq 1} \sum_{t=1}^{T-m} \sum_{g=1}^G |N_{t,g}^{T \cap G}| |N_{t+m,g}^{T \cap G}| \theta_{n,m}^{1-2/p} \rightarrow 0$;
- (c) $\frac{1}{\lambda_n} \sum_{m=M+1}^{T-1} \sum_{t=1}^{T-m} |N_t^T| |N_{t+m}^T| \theta_{n,m}^{1-2/p} \rightarrow 0$.

In the AR(1) example with $\rho \in (0, 1)$ and the triangular kernel, all three conditions are implied by ρ^m decay and the balanced panel structure.

Theorem 5 (Consistency of ET). *The following hold.*

- (a) $V_{\text{ET}}^* - V_{\text{adj}}$ is positive semidefinite.
- (b) Under Assumptions 2 and 4, and provided $\lambda_{\min}(V_{\text{ET}}^*)/\lambda_n \geq 1 + o(1)$ and $Mn/\lambda_n \rightarrow 0$, we have $(V_{\text{ET}}^*)^{-1} \widehat{V}_{\text{ET}}^* \xrightarrow{p} I_v$.
- (c) Under Assumptions 2 and 5, $V_{\text{true}}^{-1} V_{\text{adj}} \rightarrow I_v$.

Observe that (c) implies $\lambda_{\min}(V_{\text{adj}})/\lambda_n = 1 + o(1)$, so by (a) $\lambda_{\min}(V_{\text{ET}}^*)/\lambda_n \geq 1 + o(1)$. The condition $Mn/\lambda_n \rightarrow 0$ in (b) is not an additional restriction: in the balanced panel $n = GT$ and $\lambda_n \asymp T^2 G$, so $Mn/\lambda_n = M/T$, which is exactly what Assumption 4 reduces to ($M = o(T)$, satisfied by $M \asymp T^{1/3}$). Both auxiliary conditions in (b) are therefore automatically satisfied under the assumptions.

Corollary 3 (Asymptotically valid inference). *Let $\nu \in \mathbb{R}^v$ with $\|\nu\| = 1$, and suppose the conditions of Theorem 4 and of Theorem 5 hold. Then, for $S_n = \sum_{i \in N_n} \nu'(Y_{n,i} - \mathbb{E}[Y_{n,i}])$ and any $\alpha \in (0, 1)$,*

$$\limsup_{n \rightarrow \infty} \Pr \left\{ \frac{|S_n|}{\sqrt{\nu' \widehat{V}_{\text{ET}}^* \nu}} > z_{1-\alpha/2} \right\} \leq \alpha,$$

with equality if and only if $\nu'(V_{\text{ET}}^* - V_{\text{true}})\nu = o(\nu' V_{\text{true}} \nu)$.

6.4 Extension to regression

While the theoretical exposition focuses on means, the extension to the method of moments (which includes standard regressions) is straightforward. With a regression equation $Y_i = X_i' \beta + u_i$, and array $\{(X_i', u_i)\}$ satisfying ψ -dependence, as long as $\frac{1}{n} \sum_{i \in N_n} X_i X_i'$ has a minimum eigenvalue bounded away from zero, and $\{X_i u_i\}$ and $\{\text{vec}(X_i X_i')\}$ satisfy the conditions of Theorem 4, the OLS estimator $\widehat{\beta}$ is asymptotically normal and consistent. The variance of the coefficient estimator is computed by applying $\widehat{V}_{\text{ET}}^*$ to the estimated score $X_i \widehat{u}_i$ in place of $Y_{n,i}$. Standard OLS arguments then apply when limit theorems hold. Writing $X_i \widehat{u}_i = X_i u_i - X_i X_i' (\widehat{\beta} - \beta)$, the feasible estimator differs from its infeasible counterpart $\sum_{i,j} (A_n^*)_{ij} (X_i u_i) (X_j u_j)'$ only through terms linear and quadratic in $\widehat{\beta} - \beta$, which the conditions of Theorem 4 render negligible provided $\max_i \|X_i\|^2 \|\widehat{\beta} - \beta\|$ is negligible at the CLT rate.

In the numerical results that follow, we additionally augment the methods with ‘‘C2NW’’, which is a natural extension of CGM2 to cross-cluster serial correlation.

This method is conservative, but is not necessarily optimal — its details are relegated to Appendix A.

6.5 Simulation

The simulations show how the variance estimators that are robust to heterogeneous means compare with existing methods. Data are generated from the linear model

$$Y_{gt} = \beta_0 + (\beta_1 + \eta_{gt}) X_{gt} + \delta_{gt} + U_{gt}, \quad (25)$$

where the regressor and disturbance follow the CHS variance-component structure,

$$\begin{aligned} X_{gt} &= w_\alpha \alpha_g^x + w_\gamma \gamma_t^x + w_\varepsilon \varepsilon_{gt}^x, \\ U_{gt} &= w_\alpha \alpha_g^u + w_\gamma \gamma_t^u + w_\varepsilon \varepsilon_{gt}^u, \end{aligned} \quad (26)$$

with $(\beta_0, \beta_1) = (0.1, 0.1)$ and $(w_\alpha, w_\gamma, w_\varepsilon) = (0.15, 0.20, 0.15)$. The latent components $(\alpha_g^x, \alpha_g^u, \varepsilon_{gt}^x, \varepsilon_{gt}^u)$ are mutually independent $N(0, 1)$, and the common time effects (γ_t^x, γ_t^u) follow AR(1) processes with coefficient ρ .

The two heterogeneity terms are rank-one cluster-by-time interactions

$$\eta_{gt} = \kappa c_g^\eta d_t^\eta, \quad \delta_{gt} = \delta c_g^\delta d_t^\delta,$$

where the loading vectors $c_g^\bullet$ and $d_t^\bullet$ are independent, mean-zero draws. The term η_{gt} is a mean-zero random *slope*: because it is mean-zero and independent of X , the OLS estimand for β_1 remains exactly 0.1, but the score $X_{gt}u_{gt}$ inherits the component $\eta_{gt}X_{gt}^2$, which has a heterogeneous *mean* on $\mathcal{S}_{00}$. The term δ_{gt} is an additive outcome-level component omitted from the regression, contributing heterogeneous score *variance* on the same subspace. We fix $\kappa = 0.8$ and report results with the level heterogeneity switched off ($\delta = 0$) and on ($\delta = 0.8$). We hold $G = 50$, and use $T = 140$ for most designs, comparable to the dimensions of the empirical application in Section 6.6, and vary $\rho \in \{0.25, 0.50, 0.75\}$. In each of 1,000 draws we regress Y_{gt} on X_{gt} and a constant, compute standard errors by each method, and test $H_0 : \beta_1 = 0.1$ at the 5% level. Table 1 reports the empirical rejection rates.

Three patterns emerge. First, the estimators that ignore or permit limited dependence (EHW, CRg, CRt) over-reject throughout. Second, among the dependence-

Table 1: Empirical rejection rates of nominal 5% tests of $H_0 : \beta_1 = 0.1$ under the null, $G = 50, 1000$ Monte Carlo draws. Cluster $\times$ time mean heterogeneity on the two-way demeaned subspace is present when “Het. means” is Yes ($\delta = 0.8$) and absent when No. ET is the eigenvalue-truncation estimator; C2NW is the conservative CGM2 + Newey-West estimator. Under heterogeneity, CGM and CHS over-reject while ET controls size; the advantage grows with ρ and T .

row	T	rho	Het. means	EHW	CRi	CRt	CGM	CHS	C2NW	ET
(I)	140	0.25	No	0.491	0.200	0.172	0.081	0.079	0.008	0.063
(II)	140	0.50	No	0.542	0.252	0.204	0.100	0.087	0.009	0.075
(III)	140	0.75	No	0.630	0.338	0.335	0.212	0.103	0.030	0.093
(IV)	140	0.25	Yes	0.225	0.134	0.123	0.061	0.064	0.000	0.026
(V)	140	0.50	Yes	0.252	0.170	0.158	0.103	0.076	0.004	0.036
(VI)	140	0.75	Yes	0.365	0.274	0.273	0.215	0.109	0.013	0.061
(VII)	1000	0.75	No	0.734	0.133	0.510	0.104	0.058	0.031	0.057
(VIII)	1000	0.75	Yes	0.551	0.137	0.457	0.125	0.064	0.030	0.056

EHW denotes Eicker–Huber–White; CRg and CRt denote one-way cluster-robust within g and within t ; CGM denotes Cameron–Gelbach–Miller; CHS denotes Chiang–Hansen–Sasaki (as written in their paper, without subtracting within-cluster lag); C2NW denotes the conservative estimator $\hat{V}_{C2NW}$; ET denotes the eigenvalue-truncation estimator $\hat{V}_{ET}^*$.

robust estimators, CGM and CHS over-reject substantially when mean heterogeneity is present, and the distortion grows with persistence ρ , while ET controls size far better in these designs. The conservative estimator C2NW controls size in all designs, at the cost of being noticeably conservative. Third, ET uniformly has lower rejection rates than CHS, including in the designs without level heterogeneity. This reflects the within-cluster correction embedded in $\hat{V}_{ET}^*$. The gap in size distortion is modest without level heterogeneity and widens once heterogeneity is present.

There are two qualifications. The cluster-by-time heterogeneity here is serially *uncorrelated*; this is what prevents the CHS HAC lag estimator from absorbing it. While ET controls size far better than CGM and CHS under heterogeneity, there is some over-rejection in finite-sample designs, but this over-rejection disappears when we increase T to 1000, even for the hardest case with high ρ .

6.6 Empirical Application: Minimum Wages

Cengiz et al. (2019) (CDLZ) estimate the effect of the minimum wage on low-wage employment from 138 state-level increases between 1979 and 2016, in a difference-

in-differences design run on CPS microdata aggregated to state-quarter cells. Their estimator is a bunching argument applied to the whole lower tail of the wage distribution: an increase should empty the wage bins lying below the new floor and fill those at and just above it, so the employment effect is the excess jobs appearing above the floor net of the missing jobs disappearing below it. CDLZ find that the two almost exactly offset, leaving the number of low-wage jobs unchanged while the average wage of the affected workers rises, and it is the standard errors on that pair of conclusions that we revisit. We take their Table 1, column (7), which puts the bunching estimator on a plain state-by-quarter panel with $G = 51$ states and $T = 142$ quarters. Employment and wage-bill outcomes are stacked, so each state-quarter cell contributes two observations $o \in \{E, W\}$ and the regression is

$$Y_{gt}^o = \sum_{\tau=-3}^4 \alpha_{\tau}^o D_{gt}^{\tau} + \mu_g^o + \rho_t^o + \Omega_{gt}^o + u_{gt}^o, \quad (27)$$

where Y_{gt}^E is the number of jobs paying under \$15 per hour per capita, Y_{gt}^W is the corresponding wage bill, D_{gt}^{τ} indicates event year τ relative to a minimum-wage increase, and Ω_{gt}^o collects controls for small and federal increases; every fixed effect is interacted with the outcome indicator and the regression is weighted by state population. The reported quantities are all built from the five-year post-treatment contrasts $\Delta E = \frac{1}{5} \sum_{\tau=0}^4 \alpha_{\tau}^E - \alpha_{-1}^E$ and ΔW , defined analogously. Writing E for pre-treatment employment per capita, B for the share of the workforce earning below the new minimum the year before it took effect, $\bar{w}$ for the average hourly wage of those same workers and $MWPC$ for the average size of the increase, so that $E \cdot B$ is affected employment per capita and $E \cdot B \cdot \bar{w}$ the affected wage bill, column (7) reports

$$\begin{aligned} \pi_{\text{emp}} &= \frac{\Delta E}{E \cdot B}, & \pi_{\text{mw}} &= \frac{\Delta E}{E \cdot MWPC}, \\ \pi_{\text{wage}} &= \frac{\Delta W / (E \cdot B \cdot \bar{w}) - \pi_{\text{emp}}}{1 + \pi_{\text{emp}}}, & \pi_{\text{lab}} &= \frac{\pi_{\text{emp}}}{\pi_{\text{wage}}}. \end{aligned} \quad (28)$$

Dividing ΔE by B states the employment change as a fraction of the jobs the policy directly touched, and dividing it by $MWPC$ states it as an elasticity with respect to the minimum wage, so π_{emp} and π_{mw} are one estimate under two normalisations. π_{wage} is the percentage change in the average hourly wage of those same affected workers, direct effect and spillovers together, and is the only object that combines both

Table 2: Standard Errors for Various Methods: Cengiz et al. (2019), Table 1, Column (7). 51 States, 142 Quarters

	coef	EHW	CRg	CRt	CGM	CHS	C2NW	ET
% affected wages (π_{wage})	0.065	0.046	0.010	0.017	0.013	0.023	0.036	0.026
% affected employment (π_{emp})	0.027	0.025	0.028	0.027	0.029	0.036	0.058	0.043
Emp. elast. w.r.t. MW (π_{mw})	0.023	0.022	0.024	0.023	0.025	0.031	0.050	0.037
Emp. elast. w.r.t. affected wage (π_{lab})	0.410	0.604	0.421	0.359	0.441	0.530	0.804	0.614

Column labels as in Table 5. Rows are the four objects of (28); entries are standard errors in the units of that row’s coefficient. Estimated on the 14,484 stacked observations of (27); the population weights and the eight absorbed fixed-effect dimensions are partialled out and the score is aggregated to the state-quarter cell. The CRg column reproduces the published standard errors, and the small-sample corrections of the original clustered specification are applied uniformly across columns. The Bartlett bandwidth is $M = 6$ quarters by the Andrews rule.

outcome blocks, while π_{lab} is the own-wage elasticity of employment. These coefficient estimates are held fixed across variance estimators, so the columns of Table 2 differ only through $\hat{V}$ and every entry in a row carries the units of that row’s coefficient.

Mean heterogeneity and the CHS dependence structure are natural consequences of this setting. The units are the 50 states and the District of Columbia, so they can be viewed as the finite population. Using a design-based reading, as in Abadie et al. (2023), the heterogeneous potential outcomes are fixed and the only randomness is which states raised their minimum wage, when, and by how much, so the scores used in inference have heterogeneous means. The heterogeneous effects are also non-separable, so absorbing state-by-outcome and quarter-by-outcome effects does not remove the heterogeneity. Cross-state correlation over time is a pertinent issue because the institutional features of ballot initiatives, multi-year phase-in schedules and indexation formulas all transmit one national impulse to different states at different dates, so the realization of a minimum wage change in one state in a given time period is correlated with a that of different state in the next time period, even though the correlation would decay over time. Appendix C.4 further supports this correlation by empirically measuring second moments.

From Table 2, cross-cluster serial correlation increases the standard error relative to standard two-way clustering. The correction for mean heterogeneity is materially tighter than the conservative alternative. The substantive conclusions survive, in that the wage result has $t = 2.44$ under ET against 6.31 under the published standard error and so remains significant at the 5% level, while the employment result was already

indistinguishable from zero. One unusual property is that EHW is larger than the clustered standard errors in the wage row, and CGM falls below CRt there. Write $a_r = h'(Z'Z)^{-1}Z_re_r$ for the projected score of observation r , so that $h'\widehat{V}h = \sum_r a_r^2$ under EHW and $\sum_c (\sum_{r \in c} a_r)^2$ under any scheme grouping observations into sets c ; the two differ only by the cross-product $\sum_c \sum_{r \neq r'} a_r a_{r'}$. For π_{wage} and π_{lab} that cross-product is negative due to the sign pattern of h . The two quantities difference the employment and wage-bill blocks, so h loads on them with opposite signs, and the strongly positive covariance between the blocks becomes a negative product once projected. Clustering therefore subtracts from the EHW value rather than adding to it. The two quantities loading on the employment block alone, π_{emp} and π_{mw} , involve no such differencing.

7 Conclusion

This paper developed a unified theory of dependence-robust variance estimation under heterogeneous means. Every quadratic estimator $Y_n' A_n Y_n$ has expectation $\text{tr}(A_n \Sigma) + \mu' A_n \mu$. Cone duality turns this decomposition into an exact conservativeness criterion (Theorem 2), and the eigenvalue truncation W_n^+ solves several optimality criteria under some conditions (Theorem 3). The same construction doubles as a diagnostic: one-way cluster-robust estimators, and Bartlett-kernel HAC emerge as optimal, whereas two-way clustering is anticonservative under heterogeneous means and is repaired by an explicit correction. Specializing the theory resolves an open problem in the panel literature—a variance estimator simultaneously robust to heterogeneous means and cross-cluster serial correlation.

A Supporting Results

A.1 Details on Optimization Problems

In the minimal correction problem, the constraint $\|\mu\| = 1$ excludes $\mu = 0$. When $\mathcal{M}_n = \{0\}$ —the mean is known exactly, so that the demeaned mean vanishes—the supremum in (7) is over the empty set; we adopt the convention $\mathcal{R}_n(A_n) = 0$, reflecting that (C2) is vacuous and no mean-induced correction is warranted, so the plug-in $A_n = W_n$ is feasible and optimal.

Under finite fourth moments, expanding the squared error gives

$$\text{MSE}_n(A_n; P_n) = [\text{tr}((A_n - W_n)\Sigma) + \mu' A_n \mu]^2 + 2 \text{tr}((A_n \Sigma)^2) + 4 \mu' A_n \Sigma A_n \mu + \mathcal{T}_3(A_n; P_n) + \mathcal{K}_4(A_n; P_n), \quad (29)$$

where, writing $Z = Y_n - \mu$, $m_{ijk}^{(3)}(P_n) = \mathbb{E}_{P_n}[Z_i Z_j Z_k]$, and $\kappa_{ijkl}(P_n) = \mathbb{E}_{P_n}[Z_i Z_j Z_k Z_l] - \mathbb{E}[Z_i Z_j] \mathbb{E}[Z_k Z_l] - \mathbb{E}[Z_i Z_k] \mathbb{E}[Z_j Z_l] - \mathbb{E}[Z_i Z_l] \mathbb{E}[Z_j Z_k]$,

$$\mathcal{T}_3(A_n; P_n) := 4 \sum_{i,j,k} (A_n)_{ij} (A_n \mu)_k m_{ijk}^{(3)}(P_n), \quad \mathcal{K}_4(A_n; P_n) := \sum_{i,j,k,l} (A_n)_{ij} (A_n)_{kl} \kappa_{ijkl}(P_n).$$

Under Assumption 1, MSE reduces to

$$\text{MSE}_n(A_n; \mu, \Sigma) = [\text{tr}((A_n - W_n)\Sigma) + \mu' A_n \mu]^2 + 2 \text{tr}((A_n \Sigma)^2) + 4 \mu' A_n \Sigma A_n \mu, \quad (30)$$

depending on P_n only through (μ, Σ) .

Outside the environment of Theorem 3 — when $\mathcal{M}_n \neq \mathbb{R}^n$, or when the researcher declines the enlargement $\mathcal{S}_n^\oplus$ — no correction dominates pointwise. One way to select an estimator commits to how the uncertainty in (μ, Σ) is weighted. Let π be a researcher-declared probability measure on $\mathcal{M}_n \times \mathcal{S}_n$ with finite fourth moments, write $\Psi_\pi := \mathbb{E}_\pi[\Sigma + \mu \mu'] \succeq 0$, and replace the pointwise problems (9)–(10) by their π -averages,

$$\min_{A_n \in \text{Sym}_n} \int \text{MSE}_n(A_n; \mu, \Sigma) d\pi(\mu, \Sigma) \quad \text{s.t.} \quad A_n - W_n \in \mathcal{S}_n^*, \quad P_{\mathcal{M}_n} A_n P_{\mathcal{M}_n} \succeq 0, \quad (31)$$

$$\min_{A_n \in \text{Sym}_n} \text{tr}(A_n \Psi_\pi) \quad \text{s.t.} \quad A_n - W_n \in \mathcal{S}_n^*, \quad P_{\mathcal{M}_n} A_n P_{\mathcal{M}_n} \succeq 0. \quad (32)$$

Averaging names no adversary; π is irrelevant for validity, since the constraints bind whatever π is declared. All three programs are conic in A_n : the feasible set is the intersection of the closed convex cone constraint $A_n - W_n \in \mathcal{S}_n^*$ with a linear matrix inequality; (32) is linear in A_n ; (31) is convex whenever the admissible moment tensors make $\text{MSE}_n(\cdot; \mu, \Sigma)$ convex for π -almost every (μ, Σ) ; and (8) becomes $\min t$ subject to $t P_{\mathcal{M}_n} \succeq P_{\mathcal{M}_n} (A_n - W_n) P_{\mathcal{M}_n}$. Any instance can therefore be solved by an interior-point method once $(W_n, \mathcal{M}_n, \mathcal{S}_n)$ and the moment-tensor class are specified.

Two things limit the value of this route. First, where Theorem 3 applies, A_n^*

minimizes the integrand at every (μ, Σ) in scope and hence the integral for *every* π , so the averaging device is redundant exactly where it would be least controversial. Second, where it does not apply, the numerical optima are hard to use: they need not converge to the estimand they target, and in the restricted-mean regime the level objective can be minimized by a degenerate estimator. The study of its statistical properties is left for future research.

A.2 Examples for Panel Dependence

The following example shows that the CHS estimand can be anticonservative under heterogeneous means.

Example 3. Suppose we have one observation in each (g, t) intersection, and we have $m = 1$ dependence. We can compare the oracle CHS variance estimand (where $\omega(m, M) = 1$ for all $m \leq 1$ and 0 otherwise) with the true variance. As before, this example can be extended to $T \rightarrow \infty$ and $G \rightarrow \infty$. The scalar $E[Y_{n,i}]$ values are in Table 3.

Table 3: Values of scalar $E[Y_{n,i}]$

	$t = 1$	$t = 2$	$t = 3$	$t = 4$
$g = 1$	-1	-1	1	1
$g = 2$	1	1	-1	-1

Then, observe that $\sum_{i \in N_n} \sum_{j \in N_{g(i)}^G} E[Y_{n,i}] E[Y_{n,j}] = 0$, $\sum_{i \in N_n} \sum_{j \in N_{t(i)}^T} E[Y_{n,i}] E[Y_{n,j}] = 0$, and $\sum_{i \in N_n} \sum_{j \in N_{t(i),g(i)}^{T \cap G}} E[Y_{n,i}] E[Y_{n,j}] = 8$. In the serially correlated components, setting $m = 1$, $\sum_{t=1}^{T-m} E[y_t] E[y'_{t+m}] = 0$ and $\sum_{t=1}^{T-m} \sum_{g=1}^G \sum_{i \in N_{t,g}^{T \cap G}} \sum_{j \in N_{t+m,g}^{T \cap G}} E[Y_{n,i}] E[Y_{n,j}] = 2$. Then, on aggregate,

$$\begin{aligned}
& \sum_{i \in N_n} \sum_{j \in N_{g(i)}^G} E[Y_{n,i}] E[Y_{n,j}] + \sum_{i \in N_n} \sum_{j \in N_{t(i)}^T} E[Y_{n,i}] E[Y_{n,j}] - \sum_{i \in N_n} \sum_{j \in N_{t(i),g(i)}^{T \cap G}} E[Y_{n,i}] E[Y_{n,j}] \\
& + 2 \sum_{t=1}^{T-m} E[y_t] E[y_{t+m}] - 2 \sum_{t=1}^{T-m} \sum_{g=1}^G \sum_{i \in N_{t,g}^{T \cap G}} \sum_{j \in N_{t+m,g}^{T \cap G}} E[Y_{n,i}] E[Y_{n,j}] = -12 < 0, \quad (33)
\end{aligned}$$

so the standard CHS estimand is anticonservative.

The following example satisfies ψ -dependence but not the Aldous-Hoover representation.

Example 4. Suppose every firm g is observed for T time periods, each indexed by t . Data is generated from the following process. There is an innovation sequence $(\nu_{gt})_{t \in \mathbb{Z}}$ iid $N(0, 1)$ over g, t . For some $|\rho| \in (0, 1)$, $D_{gt} = \rho D_{gt-1} + \nu_{gt}$, $t \in \mathbb{Z}$, with D_{gt} stationary and processes independent across g . Hence, within a firm, $\{D_{gt}\}_t$ is an AR(1) Markov chain with parameter ρ , and across firms, $\{D_{gt}\}_g$ are independent copies of this AR(1) process.

Such a process does not permit the CHS representation. Suppose to a contradiction that D_{gt} admits a representation where $D_{gt} = f(\alpha_g, \gamma_t, \varepsilon_{gt})$ for iid $\alpha_g, \varepsilon_{gt}$, and strictly stationary γ_t . In the DGP, $\gamma_t = 0$, because if γ_t is stochastic, then there must be dependence across g units, which is ruled out in this DGP. With the resulting representation $D_{gt} = f(\alpha_g, 0, \varepsilon_{gt})$, it follows that $\text{Cov}(D_{gt}, D_{gs_1}) = \text{Cov}(D_{gt}, D_{gs_2})$ for $t \neq s_1 \neq s_2 \neq t$. However, this is untrue in the DGP as $\text{Cov}(D_{g,t}, D_{g,t-1}) = \rho/(1 - \rho^2) \neq \rho^2/(1 - \rho^2) = \text{Cov}(D_{g,t}, D_{g,t-2})$.

A.3 Variance estimators

We now define both variance estimators and their target estimands. The CHS variance estimator (excluding their EVC operation) is:

$$\begin{aligned} \hat{V}_{\text{CHS}} = & \sum_{i \in N_n} \sum_{j \in N_{g(i)}^G} Y_{n,i} Y'_{n,j} + \sum_{i \in N_n} \sum_{j \in N_{t(i)}^T} Y_{n,i} Y'_{n,j} - \sum_{i \in N_n} \sum_{j \in N_{t(i),g(i)}^{T \cap G}} Y_{n,i} Y'_{n,j} \\ & + \sum_{m=1}^M \omega(m, M) \left(\sum_{t=1}^{T-m} y_t y'_{t+m} + \sum_{t=1}^{T-m} y_{t+m} y'_t \right) \\ & - \sum_{m=1}^M \omega(m, M) \left(\sum_{t=1}^{T-m} \sum_{g=1}^G \sum_{i \in N_{t,g}^{T \cap G}} \sum_{j \in N_{t+m,g}^{T \cap G}} Y_{n,i} Y'_{n,j} + \sum_{t=1}^{T-m} \sum_{g=1}^G \sum_{i \in N_{t,g}^{T \cap G}} \sum_{j \in N_{t+m,g}^{T \cap G}} Y_{n,j} Y'_{n,i} \right). \end{aligned} \quad (34)$$

The first line is the CGM estimator, the second accounts for serial correlation across clusters, and the third adjusts for double counting. The CHS estimand V_{CHS} replaces each empirical product with its expectation.

Lemma 1 (Matrix form of the CHS estimator). *In the balanced panel with one observation per cell, $i = (g, t)$ and $\mathbb{R}^n \cong \mathbb{R}^T \otimes \mathbb{R}^G$, the CHS variance estimator (34)*

admits the quadratic-form representation

$$\hat{V}_{\text{CHS}} = Y_n' W_n^{\text{CHS}} Y_n, \quad W_n^{\text{CHS}} = J_T \otimes I_G + C_M \otimes (J_G - I_G).$$

A simpler conservative estimator extends [Davezies et al. \(2021\)](#)'s CGM2 construction to the serially correlated setting. CGM2 adds W_n^T to the CGM weight matrix to restore uniform conservativeness in the standard two-way setting (Section 5.3); its analog here adds a tractable PSD term $2 \sum_t y_t y_t'$ in the kernel-weighted block:

$$\begin{aligned} \hat{V}_{\text{C2NW}} := & \sum_{i \in N_n} \sum_{j \in N_{g(i)}^G} Y_{n,i} Y_{n,j}' + \sum_{i \in N_n} \sum_{j \in N_{t(i)}^T} Y_{n,i} Y_{n,j}' \\ & + \sum_{m=1}^M \omega(m, M) \left(\sum_{t=1}^{T-m} y_t y_{t+m}' + \sum_{t=1}^{T-m} y_{t+m} y_t' + 2 \sum_{t=1}^T y_t y_t' \right). \end{aligned} \quad (35)$$

V_{C2NW} is defined analogously by replacing empirical products with expectations. The estimator omits the $\sum_{t,g,i,j}$ double-counting line of $\hat{V}_{\text{CHS}}$, matching CHS's own implementation choice.

By construction, $\hat{V}_{\text{C2NW}}$ adds weight on the group-symmetric subspace ($v \otimes \mathbf{1}_G$ directions), while $\hat{V}_{\text{ET}}^*$ adds weight precisely on the group-demeaned subspace where the negative eigenvalues live. Both are uniformly conservative, but the optimal correction targets the pathology directly while the conservative correction overshoots.

For implementation, M is selected using the bandwidth selection procedure of [Andrews \(1991\)](#) and $\omega(\cdot)$ is the triangular (Bartlett) kernel. This choice follows CHS for comparability. Assumption 4 requires only $M = o(T)$, which the resulting bandwidths satisfy.

A.4 Consistency Details

Lemma 2 (Quadratic-form law of large numbers). *Let $\{Y_{n,i}\}$ be a triangular array of scalar random variables that is conditionally ψ -dependent given $\{\mathcal{C}_n\}$, with dependence coefficients $\{\theta_{n,s}\}$ satisfying Assumption 2(b). Let $A_n \in \text{Sym}_n$ be a (possibly random but $\mathcal{C}_n$ -measurable) symmetric weight matrix, and define*

$$\hat{V}_{A,n} := Y_n' A_n Y_n, \quad V_{A,n} := \mathbb{E}[Y_n' A_n Y_n \mid \mathcal{C}_n].$$

Suppose the following primitive conditions hold for some $p > 4$:

- (i) (Moment bound) $\sup_{n \geq 1} \max_{i \in N_n} \|Y_{n,i}\|_{\mathcal{C}_n, p} < \infty$ a.s.
- (ii) (Weight bound) There is a nonnegative sequence $\{\bar{a}_n\}$ with $\max_{i,j \in N_n} |(A_n)_{ij}| \leq \bar{a}_n$ a.s. for every n .
- (iii) (Support bandwidth) There exists a sequence $b_n \geq 0$ such that $(A_n)_{ij} = 0$ whenever $d_n(i, j) > b_n$.
- (iv) (Variance summability) $\bar{a}_n^2 \cdot n \sum_{s \geq 0} c_n(s, b_n; 2) \theta_{n,s}^{1-4/p} \rightarrow 0$ a.s.

Then $\hat{V}_{A,n} - V_{A,n} \xrightarrow{p} 0$.

The lemma is applied to weight matrices that have already been normalized by λ_n : for an unnormalized A_n with $\max_{i,j} |(A_n)_{ij}| \leq a_n$, taking A_n/λ_n in the lemma gives $\bar{a}_n = a_n/\lambda_n$ and recovers $\lambda_n^{-1}(\hat{V}_{A,n} - V_{A,n}) \xrightarrow{p} 0$ under $\frac{a_n^2}{\lambda_n^2} n \sum_{s \geq 0} c_n(s, b_n; 2) \theta_{n,s}^{1-4/p} \rightarrow 0$. Condition (ii) alone does not deliver (iv): the sum $n \sum_{s \geq 0} c_n(s, b_n; 2) \theta_{n,s}^{1-4/p}$ diverges, so the rate at which $\bar{a}_n$ vanishes is what does the work.

Remark 5. Conditions (i), (ii), and (iv) are direct adaptations of Assumption 4.1 of KMS to matrix-form weights. Condition (iii) holds whenever A_n^* inherits the support pattern of the underlying plug-in W_n , as for the block and banded cones of Section 2. It fails for the exact truncation of Corollary 2, whose correction $B^- \otimes Q_G$ is globally supported in time; that case is handled in Proposition 3 below by applying the lemma to the band-limited proxy and bounding the difference in operator norm.

We establish consistency of the eigenvalue-truncation estimator $\hat{V}_{\text{ET}}^*$ of (15) and of the conservative estimator $\hat{V}_{\text{C2NW}}$ of Appendix A.3. The argument has three layers. First, two band-limited quadratic forms—the conservative $\hat{V}_{\text{C2NW}}$ and a proxy $\hat{V}_{\text{band}}$ for the truncation—obey a law of large numbers directly through Lemma 2. Second, the exact truncation $\hat{V}_{\text{ET}}^*$, which is not band-limited, inherits consistency from the proxy $\hat{V}_{\text{band}}$ because the two are asymptotically indistinguishable. Third, the common target V_{adj} is asymptotically equivalent to V_{true} . Throughout, $\hat{V}_{\text{band}} := Y_n' A_n^{\text{band}} Y_n$ for

$$A_n^{\text{band}} := W_n^{\text{CHS}} + C_M \otimes Q_G = J_T \otimes I_G + \frac{G-1}{G} C_M \otimes J_G, \quad V_{\text{band}} := \mathbb{E}[\hat{V}_{\text{band}} \mid \mathcal{C}_n],$$

the band-limited proxy; it plays no role beyond the consistency proof.

Proposition 2 (Conservativeness). $V_{\text{ET}}^* - V_{\text{adj}}$ and $V_{\text{C2NW}} - V_{\text{adj}}$ are both positive semidefinite.

The weight matrix of $\widehat{V}_{\text{C2NW}}$, unlike those of $\widehat{V}_{\text{CHS}}$ and $\widehat{V}_{\text{ET}}^*$, is not entrywise bounded: the conservative term $2 \sum_t y_t y'_t$ sits inside the bracket summed over m , so every lag deposits on the same within-period pairs, which accumulate weight $2 \sum_{m=1}^M \omega(m, M) = M$. The lag terms are innocuous by contrast: $\omega(m, M) \sum_t y_t y'_{t+m}$ deposits on pairs at temporal distance *exactly* m , so distinct lags occupy disjoint entries and each receives a single kernel weight. Since the variance bound of Lemma 2 scales with the square of the largest entry, $\widehat{V}_{\text{C2NW}}$ requires an additional condition.

Assumption 6. $\frac{M^2 n}{\lambda_n^2} \sum_{s \geq 0} c_n(s, 0; 2) \theta_{n,s}^{1-4/p} \rightarrow 0$.

The $O(M)$ entries are confined to pairs at distance 0, so Assumption 6 evaluates $c_n(\cdot)$ at bandwidth 0 rather than M ; since $c_n(s, 0; 2) = 0$ for $s \geq 1$, the sum collapses to its $s = 0$ term. In the balanced panel $c_n(0, 0; 2) \asymp (G + T)^2$ and $\lambda_n \asymp T^2 G$, so the left-hand side is $\asymp M^2 / T^2$ — the same order as that of Assumption 4, and likewise reducing to $M = o(T)$. The two estimators $\widehat{V}_{\text{C2NW}}$ and $\widehat{V}_{\text{ET}}^*$ therefore require the same bandwidth condition in the balanced panel, and the case for $\widehat{V}_{\text{ET}}^*$ rests on the efficiency, not on weaker regularity.

Lemma 3 (Consistency of the band-limited estimators). *Suppose Assumption 2 holds and $\lambda_{\min}(V_\bullet) / \lambda_n \geq 1 + o(1)$. Then*

$$V_\bullet^{-1} \widehat{V}_\bullet \xrightarrow{p} I_v,$$

for $\bullet = \text{band}$ under Assumption 4, and for $\bullet = \text{C2NW}$ under Assumptions 4 and 6.

Proposition 3 (Consistency of the eigenvalue-truncation estimator). *Suppose Assumptions 2 and 4 hold, $\lambda_{\min}(V_{\text{ET}}^*) / \lambda_n \geq 1 + o(1)$, and $Mn / \lambda_n \rightarrow 0$. Then*

$$(V_{\text{ET}}^*)^{-1} \widehat{V}_{\text{ET}}^* \xrightarrow{p} I_v.$$

B Proofs of Theorems

Let $N_n(i; s)$ denote the set of observations that are within distance s of observation i , and let $N_n^\partial(i; s)$ denote the set of observations that are exactly at distance s from observation i :

$$N_n(i; s) := \{j \in N_n : d_n(i, j) \leq s\}; \text{ and}$$

$$N_n^\partial(i; s) := \{j \in N_n : d_n(i, j) = s\}. \quad (36)$$

For matrices A, B , the notation $A \geq B$ means $A - B$ is positive semidefinite (psd).

Proof of Theorem 1. Fix any $A_n \in \text{Sym}_n$ and any $\Sigma_0 \in \mathcal{S}_n$, $v_n \in \mathcal{M}_n$ satisfying the hypotheses. Define $\Sigma_1 := \Sigma_0 + v_n v_n' \in \mathcal{S}_n$ and consider

$$P_n : Y_n \sim N(0, \Sigma_1), \quad Q_n : Y_n \sim N(v_n, \Sigma_0).$$

Both have finite second moments, mean in $\mathcal{M}_n$ (since $0, v_n \in \mathcal{M}_n$), and covariance in $\mathcal{S}_n$, so both lie in $\mathcal{P}_n(\mathcal{M}_n, \mathcal{S}_n)$. The expectations of the quadratic estimator under each distribution are

$$\begin{aligned} \mathbb{E}_{P_n}[\widehat{V}_{A,n}] &= \text{tr}(A_n \Sigma_1) + 0 = \text{tr}(A_n \Sigma_0) + v_n' A_n v_n, \\ \mathbb{E}_{Q_n}[\widehat{V}_{A,n}] &= \text{tr}(A_n \Sigma_0) + v_n' A_n v_n, \end{aligned}$$

which are identical for every symmetric A_n . However, the target values differ:

$$\theta_n(P_n) - \theta_n(Q_n) = \text{tr}(W_n(\Sigma_1 - \Sigma_0)) = v_n' W_n v_n.$$

Let $\Delta := v_n' W_n v_n$. For any choice of A_n , since the estimator expectation is common across P_n and Q_n while the target differs by Δ , the absolute bias must satisfy

$$|\mathbb{E}_{P_n}[\widehat{V}_{A,n}] - \theta_n(P_n)| + |\mathbb{E}_{Q_n}[\widehat{V}_{A,n}] - \theta_n(Q_n)| \geq |\Delta|,$$

so $\sup_{P \in \{P_n, Q_n\}} |\mathbb{E}_P[\widehat{V}_{A,n}] - \theta_n(P)| \geq |\Delta|/2$, and hence the supremum over the full model is at least $|\Delta|/2$. □

Proof of Theorem 2. By (1), conservativeness is

$$\text{tr}((A_n - W_n)\Sigma) + \mu' A_n \mu \geq 0 \quad \text{for all } (\mu, \Sigma) \in \mathcal{M}_n \times \mathcal{S}_n.$$

Since $0 \in \mathcal{M}_n$ as $\mathcal{M}_n$ is a linear subspace of $\mathbb{R}^n$ and $0 \in \mathcal{S}_n$ as $\mathcal{S}_n$ is a closed convex cone, taking $\mu = 0$ and $\Sigma = 0$ separately decouples the joint requirement into (3)–(4). Sufficiency is immediate. □

Proof of Theorem 3. Throughout, $\mathcal{M}_n = \mathbb{R}^n$, so (C2) reads $A_n \succeq 0$ and $\mathcal{R}_n(A_n) = \sup_{\|\mu\|=1} \mu'(A_n - W_n)\mu = \lambda_{\max}(A_n - W_n)$. Write $A_{jk} := q'_{n,j} A_n q_{n,k}$ and $c_j := A_{jj}$, so that (14) reads $c_j \geq \lambda_{n,j}^+$, with equality at A_n^* .

(i). Let A_n be uniformly conservative and set $\Delta := A_n - W_n$. Two lower bounds combine. First, take a unit eigenvector u with $W_n u = \lambda_n^{\min} u$; by (C2), $u' A_n u \geq 0$, so

$$\mathcal{R}_n(A_n) \geq u' \Delta u = u' A_n u - \lambda_n^{\min} \geq -\lambda_n^{\min}.$$

Second, since $\mathcal{S}_n \neq \{0\}$ pick $\Sigma_0 \in \mathcal{S}_n \setminus \{0\}$; it is positive semidefinite and nonzero, so if $\mathcal{R}_n(A_n) < 0$, i.e. $\Delta \prec 0$, then $\text{tr}(\Delta \Sigma_0) < 0$, contradicting $\Delta \in \mathcal{S}_n^*$. Hence $\mathcal{R}_n(A_n) \geq 0$. Combining, $\mathcal{R}_n(A_n) \geq (-\lambda_n^{\min})_+$. The truncation attains this bound and is feasible: $A_n^* - W_n = W_n^- \succeq 0 \in \mathcal{S}_n^*$ gives (C1), $A_n^* = W_n^+ \succeq 0$ gives (C2), and $\mathcal{R}_n(A_n^*) = \lambda_{\max}(W_n^-) = (-\lambda_n^{\min})_+$. So A_n^* solves (8).

Parts (ii) and (iii) rest on (14) and on the following facts. First, if *either* A_n or Ψ is aligned, then

$$\text{tr}(A_n \Psi) = \sum_j c_j \psi_j, \quad (37)$$

since an aligned $\Psi = \sum_j \psi_j q_{n,j} q'_{n,j}$ gives $\text{tr}(A_n \Psi) = \sum_j \psi_j q'_{n,j} A_n q_{n,j}$, and an aligned $A_n = \sum_j c_j q_{n,j} q'_{n,j}$ gives $\text{tr}(A_n \Psi) = \sum_j c_j q'_{n,j} \Psi q_{n,j}$. Second, a positive semidefinite matrix with a zero diagonal entry has the corresponding row and column equal to zero. Further, an aligned $A_n = \sum_j c_j q_{n,j} q'_{n,j}$ lies in $\mathcal{A}_n^\oplus$ if and only if $c_j \geq \lambda_{n,j}^+$ for every j : necessity is (14), and for sufficiency, $c_j - \lambda_{n,j} \geq \lambda_{n,j}^- \geq 0$ makes $A_n - W_n$ positive semidefinite, hence a member of $\text{Sym}_n^+ \subseteq \mathcal{S}_n^*$ with nonnegative eigenbasis diagonal, while $c_j \geq \lambda_{n,j}^+ \geq 0$ gives $A_n \succeq 0$. Taking $c = \lambda_n^+$ shows $A_n^* \in \mathcal{A}_n^\oplus$.

(ii)(a). Let $A_n \in \mathcal{A}_n^\oplus$ and let $\Psi \succeq 0$, with either Ψ or A_n aligned. Then $\psi_j \geq 0$, so (37) and (14) give

$$\text{tr}(A_n \Psi) = \sum_j \psi_j c_j \geq \sum_j \psi_j \lambda_{n,j}^+ = \text{tr}(A_n^* \Psi).$$

Both sentences of (ii)(a) are this inequality, under the two alignment hypotheses in turn.

(ii)(b). Let $d : \text{Sym}_n \rightarrow \mathbb{R}^n$ be the linear map $d(\Sigma)_j := q'_{n,j} \Sigma q_{n,j}$, let $\{e_j\}$ be the standard basis of $\mathbb{R}^n$, and let $\mathcal{D} := d(\mathcal{S}_n)$, a convex cone contained in $\mathbb{R}_+^n$. We claim $e_k = d(q_{n,k} q'_{n,k}) \notin \overline{\mathcal{D}}$. Suppose otherwise, so that $d(\Sigma^{(m)}) \rightarrow e_k$ for some

$\Sigma^{(m)} \in \mathcal{S}_n$. Then $q'_{n,k} \Sigma^{(m)} q_{n,k} \rightarrow 1$, so after rescaling within the cone we may take it equal to 1, whence $\text{tr}(\Sigma^{(m)}) = \sum_j d(\Sigma^{(m)})_j \rightarrow 1$. The sequence is bounded, so along a subsequence $\Sigma^{(m)} \rightarrow \Sigma^* \in \mathcal{S}_n$ by closedness, with $\text{tr}(\Sigma^*) = q'_{n,k} \Sigma^* q_{n,k} = 1$. For a positive semidefinite matrix of unit trace, $q' \Sigma q \leq 1$ with equality only at $\Sigma = qq'$; hence $q_{n,k} q'_{n,k} = \Sigma^* \in \mathcal{S}_n$, contradicting the hypothesis.

Separating e_k from $\bar{D}$ supplies $a \in \mathbb{R}^n$ with $a' d(\Sigma) \geq 0$ for every $\Sigma \in \mathcal{S}_n$ and $a_k < 0$. The aligned matrix $D := \sum_j a_j q_{n,j} q'_{n,j}$ then satisfies $\text{tr}(D\Sigma) = a' d(\Sigma) \geq 0$, so $D \in \mathcal{S}_n^*$, while $q'_{n,k} D q_{n,k} = a_k < 0$. Fix $0 < \varepsilon < \lambda_{n,k}/|a_k|$ and set

$$A_n := W_n + \varepsilon D + \sum_{j \neq k} (\lambda_{n,j}^- + \varepsilon |a_j|) q_{n,j} q'_{n,j},$$

which is aligned with $c_k = \lambda_{n,k} + \varepsilon a_k$ and $c_j = \lambda_{n,j}^+ + \varepsilon(a_j + |a_j|)$ for $j \neq k$. It is uniformly conservative on $\mathcal{P}_n(\mathbb{R}^n, \mathcal{S}_n)$: the second and third terms lie in $\mathcal{S}_n^*$, the third because it is positive semidefinite, so $A_n - W_n \in \mathcal{S}_n^*$; and $c_j \geq 0$ for $j \neq k$ while $c_k > 0$ by the choice of ε , so $A_n \succeq 0$. But $0 < c_k < \lambda_{n,k} = \lambda_{n,k}^+$, so A_n violates the added inequality of (12) at k and, at $\Psi = q_{n,k} q'_{n,k}$ realized by $\Sigma = 0$ and $\mu = q_{n,k}$,

$$\text{tr}(A_n \Psi) = c_k < \lambda_{n,k} = \text{tr}(A_n^* \Psi).$$

(ii)(c). By (37) and (14), $\text{tr}(A_n \Psi) - \text{tr}(A_n^* \Psi) = \sum_j \psi_j (c_j - \lambda_{n,j}^+)$ is a sum of nonnegative terms, so $A_n \in \mathcal{A}_n^\oplus$ attains the minimum if and only if $c_j = \lambda_{n,j}^+$ whenever $\psi_j > 0$. If $\psi_{j_0} = 0$ for some j_0 , then $A_n^* + \varepsilon q_{n,j_0} q'_{n,j_0}$ is a distinct minimizer for every $\varepsilon > 0$, being aligned with $c_j \geq \lambda_{n,j}^+$. If instead $\psi_j > 0$ for every j , the minimizers are exactly the $A_n \in \mathcal{A}_n^\oplus$ with $c_j = \lambda_{n,j}^+$ for every j , that is, the matrices $A_n^* + E$ with $E \in \text{Sym}_n$, $q'_{n,j} E q_{n,j} = 0$ for every j , and $A_n^* + E \in \mathcal{A}_n^\oplus$; uniqueness is therefore equivalent to $E = 0$ being the only such E .

For the three sufficient conditions, let E be one of them and $A_n := A_n^* + E$. If A_n is aligned then E is aligned with zero diagonal, so $E = 0$. Otherwise, if $\lambda_{n,j} \leq 0$ then $A_{jj} = \lambda_{n,j}^+ = 0$, and $A_n \succeq 0$ forces the j -th row of $(A_{jk})_{j,k}$ to vanish, so $q'_{n,j} E q_{n,k} = 0$ for every k . Thus E is supported on the off-diagonal pairs $j \neq k$ with $\lambda_{n,j}, \lambda_{n,k} > 0$, of which there are none when W_n has at most one positive eigenvalue. In the remaining case fix such a pair, put $v := q_{n,j} + \tau q_{n,k}$, and apply (3) at $vv' \in \mathcal{S}_n$:

since $W_n^- q_{n,j} = W_n^- q_{n,k} = 0$ and E has zero eigenbasis diagonal,

$$0 \leq v'(A_n - W_n)v = v'(W_n^- + E)v = 2\tau q'_{n,j} E q_{n,k},$$

and taking $\tau = 1$ and $\tau = -1$ gives $q'_{n,j} E q_{n,k} = 0$.

(iii)(a). Assumption 1 gives (30). Write $\Sigma = \sum_j s_j q_{n,j} q'_{n,j}$ with $s \geq 0$ and $\mu = t q_{n,i}$ with $t := q'_{n,i} \mu$, so that $\psi_j = s_j + t^2 \mathbf{1}\{j = i\}$. Alignment of Σ collapses each term:

$$\text{tr}((A_n - W_n)\Sigma) = \sum_j (c_j - \lambda_{n,j}) s_j, \quad \mu' A_n \mu = t^2 c_i,$$

$$\text{tr}((A_n \Sigma)^2) = \sum_{j,k} A_{jk}^2 s_j s_k, \quad \mu' A_n \Sigma A_n \mu = t^2 \sum_j s_j A_{ji}^2.$$

Let $b(A_n)$ denote the bracketed bias term of (30). Then $b(A_n) - b(A_n^*) = \sum_j (c_j - \lambda_{n,j}^+) \psi_j \geq 0$ by (14), while $b(A_n^*) = \sum_j \lambda_{n,j}^- s_j + t^2 \lambda_{n,i}^+ \geq 0$, so $b(A_n)^2 \geq b(A_n^*)^2$. For the two variance terms, (14) and $\lambda_{n,j}^+ \geq 0$ give

$$\sum_{j,k} A_{jk}^2 s_j s_k \geq \sum_j c_j^2 s_j^2 \geq \sum_j (\lambda_{n,j}^+)^2 s_j^2, \quad t^2 \sum_j s_j A_{ji}^2 \geq t^2 s_i c_i^2 \geq t^2 s_i (\lambda_{n,i}^+)^2.$$

Each of the three terms of (30) is therefore minimized at A_n^* .

(iii)(b). At $\Sigma = 0$ and $\mu = q_{n,k}$ the point mass at μ satisfies Assumption 1, Σ is aligned, μ lies on an eigen-line, and $\theta_n = 0$, so (30) reduces to $\text{MSE}_n(A_n; \mu, \Sigma) = c_k^2$. The matrix constructed in (ii)(b) has $0 < c_k < \lambda_{n,k} = q'_{n,k} A_n^* q_{n,k}$, so its MSE is strictly smaller.

(iii)(c). If $\psi_{j_0} = 0$ for some j_0 then $s_{j_0} = 0$ and $q'_{n,j_0} \mu = 0$, and the distinct member $A_n^* + \varepsilon q_{n,j_0} q'_{n,j_0}$ of $\mathcal{A}_n^\oplus$ from (ii)(c) leaves all three terms unchanged, so the minimizer is not unique. Conversely suppose $\psi_j > 0$ for every j and that A_n ties. Equality in the bias term forces $c_j = \lambda_{n,j}^+$ for every j . Since $q'_{n,j} \mu = 0$ for $j \neq i$, one has $\psi_j = s_j > 0$ there; equality in the second variance term then forces $A_{jk} = 0$ for every $j \neq k$ with $s_j s_k > 0$, and equality in the third forces $A_{ji} = 0$ for every $j \neq i$. If $s_i > 0$ the former kills every off-diagonal entry; if $s_i = 0$, the pairs not covered by the latter are those with $j, k \neq i$, and these have $s_j s_k > 0$. In either case $A_n = A_n^*$. $\square$

Proof of Theorem 4. It suffices to verify the conditions of KMS Theorem 3.2, using notation in this setup. Our Assumption 2 is identical to KMS Assumption 2.1. $Y_{n,i}$ satisfying that condition implies that $\nu' Y_{n,i}$ also satisfies that condition. Assump-

tion 2(c)'s finite moment condition satisfies KMS Assumption 3.3. Observe that, since $\|\nu\| = 1$, we have $\sigma_n^2 \geq \lambda_n$ as the worst case σ_n^2 loads all weight on a combination of variance components that results in the smallest eigenvalue. Since $\sigma_n^2 \geq \lambda_n$, for some arbitrary constant $C > 0$,

$$\frac{n}{\sigma_n^{2+k}} \sum_{s \geq 0} c_n(s, m_n; k) \theta_{n,s}^{1-\frac{2+k}{p}} \leq \frac{Cn}{\lambda_n^{1+k/2}} \sum_{s \geq 0} c_n(s, m_n; k) \theta_{n,s}^{1-\frac{2+k}{p}} \xrightarrow{a.s.} 0, \text{ and} \quad (38)$$

$$\frac{n^2 \theta_{n,m_n}^{1-(1/p)}}{\sigma_n} \leq \frac{Cn^2 \theta_{n,m_n}^{1-(1/p)}}{\lambda_n^{1/2}} \xrightarrow{a.s.} 0. \quad (39)$$

It then remains to show that the setting in this paper satisfies the corresponding definitions in KMS. First, normalize $N_n(j; s) = \emptyset$ for $s < 0$, and observe that, when using $s = 0$,

$$\frac{1}{n} \sum_{i \in N_n} |N_n^\partial(i; 0)|^k \leq \frac{1}{n} \sum_{i \in N_n} (|N_{g(i)}^G| + |N_{t(i)}^T|)^k = \delta_n^\partial(0; k). \quad (40)$$

There are two cases to consider: when $m = 0$ and when $m \geq 1$. When $m = 0$, observe that, for all $s \geq 0$, we obtain

$$\frac{1}{n} \sum_{i \in N_n} |N_n(i; 0)|^k \leq \frac{1}{n} \sum_{i \in N_n} (|N_{g(i)}^G| + |N_{t(i)}^T|)^k = \Delta_n(s, 0; k), \quad (41)$$

so the $\delta_n^\partial(0; k)$ and $\Delta_n(s, 0; k)$ defined here place an upper bound on the corresponding definitions in KMS for $m = 0$. Now, consider the case of $m \geq 1$. The case of $s = 0$ is derived in Equation (40). When $s \geq 1$, the corresponding definitions are satisfied as

$$\frac{1}{n} \sum_{i \in N_n} |N_n^\partial(i; s)|^k = \frac{1}{n} \sum_{i \in N_n} (|N_{t(i)+s}^T| + |N_{t(i)-s}^T|)^k = \delta_n^\partial(s; k), \text{ and} \quad (42)$$

$$\begin{aligned} & \frac{1}{n} \sum_{i \in N_n} \max_{j \in N_n^\partial(i; s)} |N_n(i; m) \setminus N_n(j; s-1)|^k \\ &= \frac{1}{n} \sum_{i \in N_n} \max_{j \in N_n^\partial(i; s)} \left| \sum_{h=0}^m |N_{t(i)+h}^T| + \sum_{h=0}^m |N_{t(i)-h}^T| - \sum_{h=0}^{s-1} |N_{t(i)+h}^T| - \sum_{h=0}^{s-1} |N_{t(i)-h}^T| \right|^k \\ &= \frac{1}{n} \sum_{i \in N_n} \left(\sum_{h=s}^m |N_{t(i)+h}^T| + \sum_{h=s}^m |N_{t(i)-h}^T| \right)^k = \Delta_n(s, m; k). \end{aligned} \quad (43)$$

The expression $c_n(s, m; k) = \inf_{\alpha > 1} [\Delta_n(s, m; k\alpha)]^{1/\alpha} [\delta_n^\partial(s; \frac{\alpha}{\alpha-1})]^{1-\frac{1}{\alpha}}$ is identical to KMS. Consequently, the conditions of KMS Theorem 3.2 are satisfied under the Assumptions 2 and 3. $\square$

The following lemma is used for the Proof of Theorem 5.

Lemma 4. *Suppose Assumptions 2 and 5 hold and $Y_{n,i}$ is scalar. Then,*

$$\frac{1}{\lambda_n} (V_{true} - V_{adj}) \xrightarrow{p} 0. \quad (44)$$

Proof of Theorem 5. (a) By Corollary 2, $(W_n^{\text{CHS}})^+ - W_n^{\text{CHS}} = B^- \otimes Q_G \succeq 0$, and $(W_n^{\text{CHS}})^+ \succeq 0$ on $\mathcal{M}_n = \mathbb{R}^n$. Hence the bias decomposition

$$V_{\text{ET}}^* - V_{\text{adj}} = \text{tr}((B^- \otimes Q_G) \Sigma_n) + \mu_n' (W_n^{\text{CHS}})^+ \mu_n$$

has both terms nonnegative: the first is a trace of positive semidefinite matrices, the second is nonnegative by positive semidefiniteness of $(W_n^{\text{CHS}})^+$.

(b) This is Proposition 3 in Appendix A.4. The argument writes $\widehat{V}_{\text{ET}}^* = \widehat{V}_{\text{band}} - Y_n' D Y_n$, where $\widehat{V}_{\text{band}} = Y_n' (W_n^{\text{CHS}} + C_M \otimes Q_G) Y_n$ is a band-limited proxy consistent by the quadratic LLN of Lemma 2, and $D = (C_M - B^-) \otimes Q_G$ has $\|D\|_{\text{op}} \leq M + 1$, so $Y_n' D Y_n = O_p(Mn) = o_p(\lambda_n)$.

(c) It suffices to show $\frac{1}{\lambda_n} \nu' (V_{\text{true}} - V_{\text{adj}}) \nu = o(1)$ for any unit vector ν ; setting $\tilde{Y}_{n,i} = \nu' Y_{n,i}$, this is Lemma 4, whose conditions are immediate from Assumption 5. The discarded long-lag and boundary terms vanish relative to λ_n . $\square$

References

- ABADIE, A., S. ATHEY, G. W. IMBENS, AND J. M. WOOLDRIDGE (2020): “Sampling-based versus design-based uncertainty in regression analysis,” *Econometrica*, 88, 265–296.
- (2023): “When should you adjust standard errors for clustering?” *The Quarterly Journal of Economics*, 138, 1–35.
- ANDREWS, D. W. (1991): “Heteroskedasticity and autocorrelation consistent covariance matrix estimation,” *Econometrica: Journal of the Econometric Society*, 817–858.

- CAMERON, A. C., J. B. GELBACH, AND D. L. MILLER (2011): “Robust inference with multiway clustering,” *Journal of Business & Economic Statistics*, 29, 238–249.
- CASINI, A. (2023): “Theory of evolutionary spectra for heteroskedasticity and autocorrelation robust inference in possibly misspecified and nonstationary models,” *Journal of Econometrics*, 235, 372–392.
- CENGIZ, D., A. DUBE, A. LINDNER, AND B. ZIPPERER (2019): “The effect of minimum wages on low-wage jobs,” *The Quarterly Journal of Economics*, 134, 1405–1454.
- CHAN, K. W. (2022): “Optimal difference-based variance estimators in time series: A general framework,” *The Annals of Statistics*, 50, 1376–1400.
- CHIANG, H. D., B. E. HANSEN, AND Y. SASAKI (2024): “Standard errors for two-way clustering with serially correlated time effects,” *Review of Economics and Statistics*, 1–40.
- DAVEZIES, L., X. D’HAULTFŒUILLE, AND Y. GUYONVARCH (2021): “Empirical process results for exchangeable arrays,” *The Annals of Statistics*, 49, 845–862.
- FAMA, E. F. AND K. R. FRENCH (1993): “Common risk factors in the returns on stocks and bonds,” *Journal of Financial Economics*, 33, 3–56.
- KOJEVNIKOV, D., V. MARMER, AND K. SONG (2021): “Limit theorems for network dependent random variables,” *Journal of Econometrics*, 222, 882–908.
- NEWBY, W. AND K. WEST (1987): “A Simple, Positive Semi-definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix,” *Econometrica*, 55, 703–708.
- XU, R. AND L. YAP (2024): “Clustering with Potential Multidimensionality: Inference and Practice,” *arXiv preprint arXiv:2411.13372*.
- YAP, L. (2025): “Asymptotic theory for two-way clustering,” *Journal of Econometrics*, 249, 106001.

C Supplemental Appendix: Not for Publication

C.1 Proofs of Propositions

Proof of Proposition 1. Since $A_n^* - W_n = W_n^-$, $V_{A^*,n} = \text{tr}(W_n \Sigma) + \text{tr}(W_n^- \Sigma) + \mu' W_n^+ \mu$, which is (6); both terms are nonnegative because $W_n^-, W_n^+ \succeq 0$. For the first, $\text{tr}(W_n^- \Sigma) = \text{tr}(\Sigma^{1/2} W_n^- \Sigma^{1/2}) = 0$ with a positive semidefinite argument forces $(W_n^-)^{1/2} \Sigma^{1/2} = 0$, i.e. $W_n^- \Sigma = 0$, which is $\text{range}(\Sigma) \subseteq \ker W_n^-$. For the second, $\mu' W_n^+ \mu = 0$ with $W_n^+ \succeq 0$ forces $W_n^+ \mu = 0$. The converse is immediate. $\square$

Proof of Proposition 2. Throughout write $a_i := \mathbb{E}[Y_{n,i} \mid \mathcal{C}_n]$ and $\bar{y}_t := \mathbb{E}[y_t \mid \mathcal{C}_n] = \sum_{i \in N_t^T} a_i$. $\widehat{V}_{\text{ET}}^*$. By Corollary 2, $(W_n^{\text{CHS}})^+ - W_n^{\text{CHS}} = B^- \otimes Q_G \succeq 0$ and $(W_n^{\text{CHS}})^+ \succeq 0$, so

$$V_{\text{ET}}^* - V_{\text{adj}} = \text{tr}((B^- \otimes Q_G) \Sigma_n) + \mu'_n (W_n^{\text{CHS}})^+ \mu_n \geq 0,$$

the first term a trace of a product of positive semidefinite matrices, the second non-negative by $(W_n^{\text{CHS}})^+ \succeq 0$.

$\widehat{V}_{\text{C2NW}}$. Replacing each empirical product in $\widehat{V}_{\text{C2NW}}$ by its conditional expectation and subtracting V_{adj} of (24), the covariance parts of the two cluster double-sums and of the lag terms cancel against V_{adj} , leaving the within-cell covariance, the group/time mean outer products, and the mean part of the kernel-weighted block:

$$\begin{aligned} V_{\text{C2NW}} - V_{\text{adj}} = & \underbrace{\sum_i \sum_{j \in N_{g(i)}^G} a_i a_j'}_{\text{(I) within-group means}} + \underbrace{\sum_i \sum_{j \in N_{t(i)}^T} a_i a_j'}_{\text{(II) within-time means}} + \underbrace{\sum_i \sum_{j \in N_{t(i),g(i)}^{T \cap G}} \text{Cov}(Y_{n,i}, Y_{n,j} \mid \mathcal{C}_n)}_{\text{(III) within-cell covariance}} \\ & + \underbrace{\sum_{m=1}^M \omega(m, M) \sum_{t=1}^{T-m} (\bar{y}_t \bar{y}_{t+m}' + \bar{y}_{t+m} \bar{y}_t')}_{\text{(IV) kernel-weighted mean block}} + 2 \sum_{m=1}^M \omega(m, M) \sum_{t=1}^T \bar{y}_t \bar{y}_t'. \end{aligned}$$

Terms (I)–(III) are positive semidefinite: (I) telescopes to $\sum_g (\sum_{i \in N_g^G} a_i) (\sum_{i \in N_g^G} a_i)'$ and (II) likewise to $\sum_t (\sum_{i \in N_t^T} a_i) (\sum_{i \in N_t^T} a_i)'$, each a sum of outer products; (III) equals $\sum_{g,t} \text{Var}(\sum_{i \in N_{t,g}^{T \cap G}} Y_{n,i} \mid \mathcal{C}_n) \succeq 0$. For (IV), fix m and complete the square lag by lag. For each $t \leq T - m$,

$$\bar{y}_t \bar{y}_{t+m}' + \bar{y}_{t+m} \bar{y}_t' = (\bar{y}_t + \bar{y}_{t+m})(\bar{y}_t + \bar{y}_{t+m})' - \bar{y}_t \bar{y}_t' - \bar{y}_{t+m} \bar{y}_{t+m}',$$

so the two cross terms are a square minus the two diagonal squares. The doubled within-period term supplies those diagonal squares. For each m ,

$$2 \sum_{t=1}^T \bar{y}_t \bar{y}'_t - \sum_{t=1}^{T-m} (\bar{y}_t \bar{y}'_t + \bar{y}_{t+m} \bar{y}'_{t+m}) = \sum_{t=1}^T c_{t,m} \bar{y}_t \bar{y}'_t, \quad c_{t,m} \in \{0, 1, 2\},$$

a nonnegative-weighted sum of outer products, since each period is used as an end-point at most twice. Therefore

$$(IV) = \sum_{m=1}^M \omega(m, M) \left[\sum_{t=1}^{T-m} (\bar{y}_t + \bar{y}_{t+m})(\bar{y}_t + \bar{y}_{t+m})' + \sum_{t=1}^T c_{t,m} \bar{y}_t \bar{y}'_t \right] \succeq 0,$$

every summand a positive semidefinite outer product and $\omega(m, M) \geq 0$ for the triangular kernel. Adding (I)–(IV) gives $V_{C2NW} - V_{\text{adj}} \succeq 0$. $\square$

Proof of Proposition 3. By the Cramér–Wold device and the minimum-eigenvalue normalization, as in the proof of Lemma 3, it suffices to show $\frac{1}{\lambda_n} \nu' (\hat{V}_{\text{ET}}^* - V_{\text{ET}}^*) \nu \xrightarrow{p} 0$ for every unit vector ν ; set $\tilde{Y}_{n,i} := \nu' Y_{n,i}$, a scalar ψ -dependent array with the dependence coefficients of Y_n .

Decomposition. The proxy and the exact estimator differ by $D := (C_M - B^-) \otimes Q_G$, since $\hat{V}_{\text{band}} - \hat{V}_{\text{ET}}^* = Y_n' ((C_M - B^-) \otimes Q_G) Y_n$. Writing $\hat{\Delta} := \tilde{Y}_n' D \tilde{Y}_n$ and $\Delta := \mathbb{E}[\hat{\Delta} \mid \mathcal{C}_n]$,

$$\nu' (\hat{V}_{\text{ET}}^* - V_{\text{ET}}^*) \nu = \nu' (\hat{V}_{\text{band}} - V_{\text{band}}) \nu - (\hat{\Delta} - \Delta).$$

By Lemma 3 the first term is $o_p(\lambda_n)$, so it remains to bound $\hat{\Delta} - \Delta$.

Operator-norm bound on D . Both C_M and $B^- = (C_M - J_T)_+$ are positive semidefinite. Since the spectral window of C_M is the Fejér kernel, $\lambda_{\max}(C_M) \leq f_{\text{Fej}}(0) = M+1$; and as $-J_T \preceq 0$, $\lambda_{\max}(B^-) = (\lambda_{\max}(C_M - J_T))_+ \leq \lambda_{\max}(C_M) \leq M+1$. Hence $\lambda_{\max}(C_M - B^-) \leq \lambda_{\max}(C_M) \leq M+1$ and $\lambda_{\min}(C_M - B^-) \geq -\lambda_{\max}(B^-) \geq -(M+1)$, so $\|C_M - B^-\|_{\text{op}} \leq M+1$. As $\|Q_G\|_{\text{op}} = 1$,

$$\|D\|_{\text{op}} = \|C_M - B^-\|_{\text{op}} \|Q_G\|_{\text{op}} \leq M+1.$$

Conclusion. By the variational characterization of the operator norm,

$$|\hat{\Delta}| = |\tilde{Y}_n' D \tilde{Y}_n| \leq \|D\|_{\text{op}} \|\tilde{Y}_n\|^2 \leq (M+1) \sum_{i \in N_n} \tilde{Y}_{n,i}^2.$$

By Assumption 2(c), $\mathbb{E}[\tilde{Y}_{n,i}^2 \mid \mathcal{C}_n] \leq \bar{c}$ a.s. uniformly in i, n , so $\mathbb{E}[|\hat{\Delta}| \mid \mathcal{C}_n] \leq (M+1)\bar{c}n = O(Mn)$ a.s. This gives both $|\Delta| \leq \mathbb{E}[|\hat{\Delta}| \mid \mathcal{C}_n] = O(Mn)$ and, by the conditional Markov inequality, $\hat{\Delta} = O_p(Mn)$. Therefore

$$\frac{|\hat{\Delta} - \Delta|}{\lambda_n} \leq \frac{|\hat{\Delta}| + |\Delta|}{\lambda_n} = O_p\left(\frac{Mn}{\lambda_n}\right) = o_p(1),$$

using $Mn/\lambda_n \rightarrow 0$. Combining the two displays gives $\frac{1}{\lambda_n}\nu'(\hat{V}_{\text{ET}}^* - V_{\text{ET}}^*)\nu \xrightarrow{p} 0$ for each fixed unit ν , the unconditional convergence following by dominated convergence as in Lemma 2. Exactly as in the proof of Lemma 3, in fixed dimension v this per-direction bound yields $\lambda_n^{-1}\|\hat{V}_{\text{ET}}^* - V_{\text{ET}}^*\|_{\text{op}} \xrightarrow{p} 0$, whence $(V_{\text{ET}}^*)^{-1}\hat{V}_{\text{ET}}^* \xrightarrow{p} I_v$ using $\|(V_{\text{ET}}^*)^{-1}\|_{\text{op}} \leq \lambda_n^{-1}(1 + o(1))$ from the hypothesis $\lambda_{\min}(V_{\text{ET}}^*)/\lambda_n \geq 1 + o(1)$. $\square$

C.2 Proofs of Lemmas

Proof of Lemma 1. Read $\hat{V}_{\text{CHS}} = \sum_{i,j} (W_n^{\text{CHS}})_{ij} Y_{n,i} Y'_{n,j}$, so that each line of (34) is the quadratic form of a weight matrix on the index pair $((t, g), (s, h))$. With one observation per cell, $N_{g(i)}^G = \{(g, s)\}_s$, $N_{t(i)}^T = \{(h, t)\}_h$, $N_{t(i), g(i)}^{T \cap G} = \{(g, t)\}$, and $y_t = \sum_h Y_{(h,t)}$, so the five lines map to

$$\begin{aligned} \text{within-group} & \sum_i \sum_{j \in N_{g(i)}^G} & \mapsto & J_T \otimes I_G, \\ \text{within-time} & \sum_i \sum_{j \in N_{t(i)}^T} & \mapsto & I_T \otimes J_G, \\ \text{within-cell} & - \sum_i \sum_{j \in N_{t(i), g(i)}^{T \cap G}} & \mapsto & -I_T \otimes I_G, \\ \text{aggregated lag} & \sum_{m \geq 1} \omega(m, M) \sum_t (y_t y'_{t+m} + y_{t+m} y'_t) & \mapsto & (C_M - I_T) \otimes J_G, \\ \text{double-count} & - \sum_{m \geq 1} \omega(m, M) \sum_t \sum_g (Y_{(g,t)} Y'_{(g,t+m)} + \text{tr.}) & \mapsto & -(C_M - I_T) \otimes I_G, \end{aligned}$$

using $(C_M)_{ts} = \omega(|t-s|, M)$ and $\omega(0, M) = 1$. Summing the five contributions and collecting the $\otimes J_G$ and $\otimes I_G$ blocks,

$$\underbrace{[I_T + (C_M - I_T)]}_{= C_M} \otimes J_G + \underbrace{[J_T - I_T - (C_M - I_T)]}_{= J_T - C_M} \otimes I_G = C_M \otimes J_G + (J_T - C_M) \otimes I_G,$$

which rearranges to $J_T \otimes I_G + C_M \otimes (J_G - I_G) = W_n^{\text{CHS}}$. $\square$

Proof of Lemma 2. We adapt the proof of Proposition 4.1 of [Kojevnikov et al. \(2021\)](#),

translating from the HAC weight notation $\omega_n(s)$ used there to the matrix entries $(A_n)_{ij}$ used here. Throughout, all probabilistic statements are conditional on $\mathcal{C}_n$; the unconditional statement follows by the dominated convergence theorem applied to the conditional probabilities, since they are bounded by 1 and converge to 0 a.s.

Step 1: Centering. Define $z_{ij} := Y_{n,i}Y_{n,j} - \mathbb{E}[Y_{n,i}Y_{n,j} \mid \mathcal{C}_n]$, so that $\mathbb{E}[z_{ij} \mid \mathcal{C}_n] = 0$. Then $\hat{V}_{A,n} - V_{A,n} = \sum_{i \in N_n} \sum_{j \in N_n} (A_n)_{ij} z_{ij}$. By condition (iii), the inner sum is restricted to pairs (i, j) with $d_n(i, j) \leq b_n$:

$$\hat{V}_{A,n} - V_{A,n} = \sum_{i \in N_n} \sum_{j \in N_n: d_n(i,j) \leq b_n} (A_n)_{ij} z_{ij}.$$

Step 2: Conditional variance bound. We compute

$$\text{Var}\left(\hat{V}_{A,n} - V_{A,n} \mid \mathcal{C}_n\right) = \sum_{(i,j,k,l)} (A_n)_{ij} (A_n)_{kl} \mathbb{E}[z_{ij} z_{kl} \mid \mathcal{C}_n],$$

where the sum runs over quadruples $(i, j, k, l) \in N_n^4$ with $d_n(i, j) \leq b_n$ and $d_n(k, l) \leq b_n$. By condition (ii), $|(A_n)_{ij} (A_n)_{kl}| \leq \bar{a}_n^2$, so by further grouping the quadruples by the distance between the pairs $\{i, j\}$ and $\{k, l\}$:

$$\mathcal{H}_n(s, b_n) := \{(i, j, k, l) \in N_n^4 : d_n(i, j) \leq b_n, d_n(k, l) \leq b_n, d_n(\{i, j\}, \{k, l\}) = s\},$$

we obtain

$$\text{Var}\left(\hat{V}_{A,n} - V_{A,n} \mid \mathcal{C}_n\right) \leq \bar{a}_n^2 \sum_{s \geq 0} \sum_{(i,j,k,l) \in \mathcal{H}_n(s, b_n)} |\mathbb{E}[z_{ij} z_{kl} \mid \mathcal{C}_n]|.$$

Step 3: Bounding individual covariances. The variables z_{ij} do not involve A_n , so the bounds here are free of $\bar{a}_n$. For $s \geq 1$, the pairs $\{i, j\}$ and $\{k, l\}$ are at distance s , so by KMS Theorem A.1 (combined with Assumption 2(a) and condition (i) above with $p > 4$),

$$|\mathbb{E}[z_{ij} z_{kl} \mid \mathcal{C}_n]| \leq C_1 \cdot \theta_{n,s}^{1-4/p} \quad \text{a.s.},$$

for a constant $C_1 > 0$ depending only on p and the moment bound in (i). For $s = 0$, the bound is uniform: $|\mathbb{E}[z_{ij} z_{kl} \mid \mathcal{C}_n]| \leq C_1$ by Cauchy–Schwarz applied to the conditional fourth moment, which is finite by condition (i).

Step 4: Counting quadruples. By inequality (A.13) of KMS, $|\mathcal{H}_n(s, b_n)| \leq 4n c_n(s, b_n; 2)$, where $c_n(s, b_n; 2)$ is defined in equation (11) of the paper.

Step 5: Combining. Substituting Steps 3 and 4 into the variance bound from Step 2, condition (iv) yields

$$\text{Var}\left(\hat{V}_{A,n} - V_{A,n} \mid \mathcal{C}_n\right) \leq 4C_1 \bar{a}_n^2 \cdot n \sum_{s \geq 0} c_n(s, b_n; 2) \theta_{n,s}^{1-4/p} \xrightarrow{\text{a.s.}} 0.$$

By Chebyshev's inequality, for any $\varepsilon > 0$,

$$\mathbb{P}\left(|\hat{V}_{A,n} - V_{A,n}| > \varepsilon \mid \mathcal{C}_n\right) \leq \frac{1}{\varepsilon^2} \text{Var}\left(\hat{V}_{A,n} - V_{A,n} \mid \mathcal{C}_n\right) \xrightarrow{\text{a.s.}} 0.$$

Taking unconditional expectations and applying dominated convergence (the conditional probabilities are bounded by 1), $\mathbb{P}\left(|\hat{V}_{A,n} - V_{A,n}| > \varepsilon\right) \rightarrow 0$. $\square$

Proof of Lemma 3. By Slutsky's lemma and the Cramér–Wold device it suffices to show $\frac{1}{\lambda_n} \nu'(\hat{V}_\bullet - V_\bullet) \nu = o_p(1)$ for unit vector ν ; set $\tilde{Y}_{n,i} := \nu' Y_{n,i}$, a scalar ψ -dependent array with the dependence coefficients of Y_n . Using $2 \sum_{m=1}^M \omega(m, M) = M$, decompose

$$A_n^{\text{C2NW}} = \underbrace{J_T \otimes I_G + C_M \otimes J_G}_{=: A_n^{\text{lag}}} + \underbrace{M I_T \otimes J_G}_{=: A_n^0},$$

and apply Lemma 2 to each of $A_n^{\text{band}}/\lambda_n$, $A_n^{\text{lag}}/\lambda_n$, and A_n^0/λ_n ; the conclusion for $\bullet = \text{C2NW}$ follows by summing the two latter limits. We verify the conditions for $\bullet \in \{\text{C2NW}, \text{band}\}$.

- (i) Moment bound: from Assumption 2(c).
- (ii) Weight bound: the entries of $A_n^{\text{band}} = W_n^{\text{CHS}} + C_M \otimes Q_G$ and of A_n^{lag} are bounded by a constant, so $\bar{a}_n \asymp \lambda_n^{-1}$; the entries of A_n^0 equal M , so $\bar{a}_n \asymp M/\lambda_n$.
- (iii) Support bandwidth: $b_n = M$ for A_n^{band} and A_n^{lag} , whose nonzero entries lie on pairs (i, j) with $g(i) = g(j)$, $t(i) = t(j)$, or $|t(i) - t(j)| \leq M$; and $b_n = 0$ for A_n^0 , whose nonzero entries lie only on pairs with $t(i) = t(j)$.
- (iv) Variance summability: for A_n^{band} and A_n^{lag} , condition (iv) of Lemma 2 reads, up to a constant, $\frac{n}{\lambda_n^2} \sum_{s \geq 0} c_n(s, M; 2) \theta_{n,s}^{1-4/p} \rightarrow 0$, which is Assumption 4. For A_n^0 it reads $\frac{M^2 n}{\lambda_n^2} \sum_{s \geq 0} c_n(s, 0; 2) \theta_{n,s}^{1-4/p} \rightarrow 0$, which is Assumption 6.

Lemma 2 then gives, for each fixed unit ν , $\frac{1}{\lambda_n} \nu'(\hat{V}_\bullet - V_\bullet) \nu \xrightarrow{p} 0$. Because v is fixed, evaluating this at the coordinate directions e_a and the pairwise sums $e_a + e_b$ controls

every entry of the symmetric $v \times v$ matrix $\widehat{V}_\bullet - V_\bullet$, so $\lambda_n^{-1} \|\widehat{V}_\bullet - V_\bullet\|_{\text{op}} \xrightarrow{p} 0$. Writing $V_\bullet^{-1} \widehat{V}_\bullet - I_v = V_\bullet^{-1} (\widehat{V}_\bullet - V_\bullet)$ and using $\|V_\bullet^{-1}\|_{\text{op}} = \lambda_{\min}(V_\bullet)^{-1} \leq \lambda_n^{-1} (1 + o(1))$ from the hypothesis,

$$\|V_\bullet^{-1} \widehat{V}_\bullet - I_v\|_{\text{op}} \leq \lambda_{\min}(V_\bullet)^{-1} \|\widehat{V}_\bullet - V_\bullet\|_{\text{op}} \xrightarrow{p} 0.$$

□

Proof of Lemma 4. Let $\vartheta_1, \vartheta_2, \vartheta_3$ denote arbitrary positive and finite constants.

$$\begin{aligned} V_{adj} - V_{true} &= \sum_{m=1}^M \sum_{t=1}^{T-m} (\omega(m, M) - 1) (\text{Cov}(y_t, y_{t+m}) + \text{Cov}(y_{t+m}, y_t)) \\ &+ \sum_{m \geq 1} \sum_{t=1}^{T-m} \sum_{g=1}^G \sum_{i \in N_{t,g}^{T \cap G}} \sum_{j \in N_{t+m,g}^{T \cap G}} (\text{Cov}(Y_{n,i}, Y_{n,j}) + \text{Cov}(Y_{n,j}, Y_{n,i})) \\ &+ \sum_{m=M+1}^{T-1} \sum_{t=1}^{T-m} \text{Cov}(y_t, y_{t+m}) + \sum_{m=M+1}^{T-1} \sum_{t=1}^{T-m} \text{Cov}(y_{t+m}, y_t). \end{aligned} \quad (45)$$

By applying Theorem A1 from KMS, for $d_n(i, j) = s \geq 1$, we obtain $|\text{Cov}(Y_{n,i}, Y_{n,j})| \leq \vartheta_2 \theta_{n,s}^{1-\frac{2}{p}}$, where $\vartheta_2 = C(\mu \vee 1)^2 \bar{\theta}$ for some constant $C > 0$. Then,

$$\begin{aligned} &\left| \sum_{m=1}^M \sum_{t=1}^{T-m} (\omega(m, M) - 1) \text{Cov}(y_t, y_{t+m}) \right| \\ &\leq \sum_{m=1}^M \sum_{t=1}^{T-m} |\omega(m, M) - 1| \sum_{i \in N_t^T} \sum_{j \in N_{t+m}^T} |\text{Cov}(Y_{n,i}, Y_{n,j})| \\ &\leq \sum_{m=1}^M |\omega(m, M) - 1| \sum_{t=1}^T \sum_{i \in N_t^T} |N_{t(i)+m}^T| \vartheta_2 \theta_{n,m}^{1-\frac{2}{p}} \\ &\leq \sum_{m=1}^M |\omega(m, M) - 1| \sum_{i \in N_n} |N_{t(i)+m}^T| \vartheta_2 \theta_{n,m}^{1-\frac{2}{p}} \leq \vartheta_2 n \sum_{m=1}^M |\omega(m, M) - 1| \delta_n^\partial(m; 1) \theta_{n,m}^{1-\frac{2}{p}}, \end{aligned} \quad (46)$$

so it suffices for this scaled object to converge.

Now, we show that $\frac{1}{\lambda_n} \sum_{m \geq 1} \sum_{t=1}^{T-m} \sum_{g=1}^G \sum_{i \in N_{t,g}^{T \cap G}} \sum_{j \in N_{t+m,g}^{T \cap G}} \text{Cov}(Y_{n,i}, Y_{n,j})$ is asymp-

totically negligible:

$$\sum_{m \geq 1} \sum_{t=1}^{T-m} \sum_{g=1}^G \sum_{i \in N_{t,g}^{T \cap G}} \sum_{j \in N_{t+m,g}^{T \cap G}} |\text{Cov}(Y_{n,i}, Y_{n,j})| \leq \sum_{m \geq 1} \sum_{t=1}^{T-m} \sum_{g=1}^G |N_{t,g}^{T \cap G}| |N_{t+m,g}^{T \cap G}| \vartheta_2 \theta_{n,m}^{1-\frac{2}{p}}. \quad (47)$$

The object converges to zero after scaling due to the assumption used.

Finally, we show $\frac{1}{\lambda_n} \sum_{m=M+1}^{T-1} \sum_{t=1}^{T-m} \text{Cov}(y_t, y_{t+m})$ is asymptotically negligible:

$$\sum_{m=M+1}^{T-1} \sum_{t=1}^{T-m} |\text{Cov}(y_t, y_{t+m})| \leq \sum_{m=M+1}^{T-1} \sum_{t=1}^{T-m} |N_t^T| |N_{t+m}^T| \vartheta_3 \theta_{n,m}^{1-\frac{2}{p}}. \quad (48)$$

The object converges to zero after scaling due to the assumption used. $\square$

C.3 Proof of Corollaries

Proof of Corollary 1. Write $P_G := J_G/G$, $Q_G := I_G - P_G$ and P_T, Q_T analogously; these are orthogonal idempotents on each factor. Substituting $J_G = GP_G$, $J_T = TP_T$, $I = P + Q$ into $W_n^{2\text{-way}}$ and collecting terms in the commuting blocks $\{P_T, Q_T\} \otimes \{P_G, Q_G\}$,

$$W_n^{2\text{-way}} = (G+T-1) P_T \otimes P_G + (G-1) Q_T \otimes P_G + (T-1) P_T \otimes Q_G - Q_T \otimes Q_G.$$

The coefficients are the eigenvalues; the only negative one is -1 , on $Q_T \otimes Q_G$ (nonempty iff $G, T \geq 2$). The positive part zeros this block, so $A_n^\star = W_n^{2\text{-way}} + Q_T \otimes Q_G$. Finally $Q_T \otimes Q_G = (I_T - P_T) \otimes (I_G - P_G) = I_n - W_n^G/T - W_n^T/G + J_n/n = P_{00}$ by inclusion-exclusion, with the stated quadratic form. $\square$

Proof of Corollary 2. Decompose $I_G = P_G + Q_G$ and $J_G - I_G = (G-1)P_G - Q_G$, and substitute into W_n^{CHS} . Since P_G, Q_G are orthogonal idempotent,

$$W_n^{\text{CHS}} = \underbrace{[J_T + (G-1)C_M]}_{=:A} \otimes P_G + \underbrace{[J_T - C_M]}_{=:B} \otimes Q_G,$$

an orthogonal direct sum over $\mathbb{R}^T \otimes \text{range}(P_G)$ and $\mathbb{R}^T \otimes \text{range}(Q_G)$, so the positive part acts blockwise: $(W_n^{\text{CHS}})^+ = A^+ \otimes P_G + B^+ \otimes Q_G$. Both J_T and C_M are PSD (the latter a Fejér kernel), so $A = J_T + (G-1)C_M \succeq 0$ and $A^+ = A$. For the second block,

$B^+ = B + B^-$ with $B^- = (C_M - J_T)_+ \succeq 0$, giving $B^+ \otimes Q_G = B \otimes Q_G + B^- \otimes Q_G$. Summing, $A_n^* = W_n^{\text{CHS}} + B^- \otimes Q_G$. Since Q_G acts on the cluster index to produce $\check{Y}$, the quadratic form is $Y_n' W_n^{\text{CHS}} Y_n + \sum_g \check{Y}_g' B^- \check{Y}_g$. $\square$

Proof of Corollary 3. Write $\sigma_n^2 := \text{Var}(S_n) = \nu' V_{\text{true}} \nu$ and $\phi_n := \nu' V_{\text{true}} \nu / \nu' V_{\text{ET}}^* \nu$; by Theorem 5(a),(c), $\nu' V_{\text{ET}}^* \nu \geq \nu' V_{\text{adj}} \nu = \nu' V_{\text{true}} \nu (1 + o(1))$, so $\phi_n \leq 1 + o(1)$. By Theorem 5(b), $\nu' \widehat{V}_{\text{ET}}^* \nu = \nu' V_{\text{ET}}^* \nu (1 + o_p(1))$, hence

$$r_n^2 := \frac{\sigma_n^2}{\nu' \widehat{V}_{\text{ET}}^* \nu} = \phi_n \frac{\nu' V_{\text{ET}}^* \nu}{\nu' \widehat{V}_{\text{ET}}^* \nu} = \phi_n (1 + o_p(1)) \leq 1 + o_p(1).$$

Since $|S_n| / \sqrt{\nu' \widehat{V}_{\text{ET}}^* \nu} = (|S_n| / \sigma_n) r_n$ and Theorem 4 gives $|S_n| / \sigma_n \xrightarrow{d} |Z|$, $Z \sim N(0, 1)$, fix $\varepsilon > 0$. On $\{r_n \leq 1 + \varepsilon\}$, $\{(|S_n| / \sigma_n) r_n > z_{1-\alpha/2}\} \subseteq \{|S_n| / \sigma_n > z_{1-\alpha/2} / (1 + \varepsilon)\}$, so

$$\Pr\left\{\frac{|S_n|}{\sqrt{\nu' \widehat{V}_{\text{ET}}^* \nu}} > z_{1-\alpha/2}\right\} \leq \Pr\left\{\frac{|S_n|}{\sigma_n} > \frac{z_{1-\alpha/2}}{1+\varepsilon}\right\} + \Pr\{r_n > 1 + \varepsilon\}.$$

The second term vanishes since $r_n \leq 1 + o_p(1)$; the first, $z_{1-\alpha/2} / (1 + \varepsilon)$ being a continuity point of the law of $|Z|$, converges to $2\Phi(-z_{1-\alpha/2} / (1 + \varepsilon))$. Taking $\limsup_n$ and then $\varepsilon \downarrow 0$,

$$\limsup_{n \rightarrow \infty} \Pr\left\{\frac{|S_n|}{\sqrt{\nu' \widehat{V}_{\text{ET}}^* \nu}} > z_{1-\alpha/2}\right\} \leq 2\Phi(-z_{1-\alpha/2}) = \alpha.$$

For the equality claim, $r_n^2 = \phi_n(1 + o_p(1))$ with $\phi_n \leq 1 + o(1)$ deterministic, so the limsup attains α iff $\phi_n \rightarrow 1$; by (a) and (c) this holds iff $\nu'(V_{\text{ET}}^* - V_{\text{true}})\nu = o(\nu' V_{\text{true}} \nu)$. $\square$

C.4 Cross-Cluster Dependence in Minimum Wage

The main text motivates cross-cluster dependence from the way minimum-wage increases are legislated, which is a claim about the assignment process. This subsection turns its attention to the score that affects the variance. Write $a_{gt} = h'(Z'Z)^{-1} \sum_{r \in (g,t)} Z_r e_r$ for the cell-aggregated score projected onto the delta-method contrast for π_{wage} in (28), so that each a_{gt} multiplies a residualised regressor by a residual. A national impulse moves how hard an increase bites, which generates an interaction in (g, t) rather than the sum of a state term and a quarter term, and the fixed effects that remove the additive channel leave it in place.

Partition the double sum over cell pairs in $h'\widehat{V}h$ into the four exhaustive and mutually exclusive sets

$$Q_{\text{own}} = \sum_{g,t} a_{gt}^2, \quad Q_g = \sum_g \sum_{t \neq s} a_{gt} a_{gs}, \quad Q_t = \sum_t \sum_{g \neq h} a_{gt} a_{ht}, \quad Q_{\times} = \sum_{g \neq h} \sum_{t \neq s} k_M(t-s) a_{gt} a_{hs},$$

where k_M is the Bartlett kernel at the bandwidth of Table 2. The first three are what the CGM weight matrix retains and $Q_{\times}$ is what it sets to zero, so the size of $Q_{\times}$ is the whole of the question. Decomposing $Q_{\times}$ by the separation $m = |t - s|$ between the two quarters gives the profile in Table 4, normalized throughout by Q_{own} . Two components enter these sums. Since $\mathbb{E}[a_{gt} a_{hs}] = \text{Cov}(a_{gt}, a_{hs}) + \mu_{gt} \mu_{hs}$, what is measured is the cross-cluster block of the estimated meat rather than a covariance alone, and under the design-based reading of Section 6.6 the second term is not a nuisance but the mean heterogeneity that motivates $\mathcal{M}_n \neq \{0\}$.

Table 4: Cross-state dependence in the score for π_{wage} , by quarter separation. $G = 51$ states; 149-quarter calendar grid, 142 observed.

m	$\rho_{\times}(m)$	$\kappa(m)$	$k_M(m) \kappa(m)$	$\sum_{j=1}^m k_M(j) \kappa(j)$
<i>Retained by CGM</i>				
0	0.0049	0.245	0.245	—
<i>Set to zero by CGM</i>				
1	0.0039	0.394	0.338	0.338
2	0.0010	0.095	0.068	0.406
3	0.0006	0.057	0.033	0.439
4	−0.0013	−0.125	−0.054	0.385
5	−0.0002	−0.024	−0.007	0.378
6	0.0021	0.212	0.030	0.408
<i>Beyond the bandwidth</i>				
7	0.0018	0.185	0.000	0.408
8	0.0007	0.075	0.000	0.408

Computed from the cell-aggregated projected score a_{gt} . $\rho_{\times}(m)$ is the one-sided average pairwise cross-state lag- m correlation and $\kappa(m) = c_m(G - 1)\rho_{\times}(m)$ is the contribution of separation m to the variance of the sum, relative to Q_{own} , with $c_0 = 1$ and $c_m = 2$ for $m \geq 1$. The factor c_m arises because the pair set $\{t \neq s\}$ contains both $s = t + m$ and $s = t - m$, whose contributions coincide once the state pairs are ordered. k_M is the Bartlett weight at $M = 6$ quarters, so $k_M(m) = 0$ for $m \geq 7$. The final column cumulates from $m = 1$; its terminal value is $Q_{\times}/Q_{\text{own}}$, and the $m = 0$ row is excluded from it because $\kappa(0) = Q_t/Q_{\text{own}}$ belongs to the set CGM retains. Cells with no observation contribute zero to every pair; the seven quarters removed by the 1994–95 CPS redesign are among them.

The final column is the object of interest. The terms CGM sets to zero cumulate to 0.408 of the own-cell term, showing how cross-cluster correlations are not negligible. The pairwise correlations in the first column are tiny: each cell has $G - 1 = 50$ cross-state partners at every separation, so what the variance of the sum sees is κ , but a correlation of four thousandths becomes a contribution of the same order as the terms CGM keeps. The last two rows show that the bandwidth is not generous: $\kappa(7) = 0.185$ is comparable to the retained $\kappa(0)$ and the kernel discards it, so 0.408 understates the dependence actually present in the score.

C.5 Empirical Application: Fama-French Model

The new variance estimator is applied to a panel of industry portfolios across time, following the application used in CHS. The Fama–French three-factor model ([Fama and French, 1993](#)) can be written as:

$$R_{gt} - R_{ft} = \beta_1(R_{Mt} - R_{ft}) + \beta_2SMB_{gt} + \beta_3HML_{gt} + e_{gt}, \quad (49)$$

where $R_{gt} - R_{ft}$ is the excess return as R_{gt} denotes the total return of portfolio g in month t , and R_{ft} denotes the risk-free rate.

I use the industry portfolio dataset from CHS containing 44 industry portfolios across 119 months. Following CHS, I use a within-transformation for both Y and $X = (R_M - R_f, SMB, HML)$ to remove additive fixed effects. Inference on coefficients relies on a CLT applied to $\sum_{g,t} \tilde{X}_{gt} \tilde{e}_{gt}$.

Table 5: Standard Errors for Various Methods: 44 Industry Portfolios, 119 Periods

	coef	EHW	CRg	CRt	CGM	CHS	C2NW	ET
MKT	0.959	0.022	0.055	0.031	0.059	0.059	0.076	0.062
SMB	0.076	0.029	0.035	0.041	0.045	0.053	0.083	0.059
HML	0.358	0.030	0.066	0.049	0.076	0.080	0.109	0.084

Estimates of the factor coefficients with various standard error estimates. EHW denotes Eicker–Huber–White; CRg denotes one-way cluster-robust within g ; CRt denotes one-way cluster-robust within t ; CGM denotes Cameron–Gelbach–Miller; CHS denotes Chiang–Hansen–Sasaki; C2NW denotes the conservative estimator $\hat{V}_{C2NW}$; ET denotes the optimal estimator $\hat{V}_{ET}^*$.

From Table 5, standard errors estimated using ET are higher than the standard errors estimated using CHS. However, the standard errors are not unreasonably large: we retain the statistical significance of the coefficient on HML at the 5% level, while the statistical significance of SMB can now be called into question, as CHS already show. Cross-cluster serial correlation is empirically important: if serial correlation were negligible, we would expect the CHS standard errors to be similar to CGM, which is not the case for two of the coefficients. We verify that ET meaningfully decreases the standard errors relative to the more conservative C2NW.